

# The i.i.d. Berry–Esseen constant is at most 0.423: a machine-checked proof

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## Abstract

Let  $C_{\text{BE}}$  be the least constant such that  $\sup_x |F_n(x) - \Phi(x)| \leq C_{\text{BE}} \beta_3 / \sqrt{n}$  for every centered, variance-one law with finite third absolute moment  $\beta_3$  and every sample size  $n \geq 1$ , where  $F_n$  is the distribution function of the normalized sum of  $n$  independent copies and  $\Phi$  is the standard normal distribution function. We describe a complete proof of  $C_{\text{BE}} \leq 0.423$  which has been formalized in Lean 4 on top of Mathlib, in the module tree `BerryEsseen/Upper0423/` of the accompanying repository, and whose final statement is

`BerryEsseen.Upper0423.berryEsseenConstant_le_0423`  
: `BerryEsseen.berryEsseenConstant`  $\leq 0.423$

The proof combines the Prawitz smoothing inequality with a family of pointwise characteristic-function estimates in the three moment coordinates  $\beta = \mathbb{E}|X|^3$ ,  $\eta = \frac{1}{2}\mathbb{E}|X - X'|^3 - \beta$  and  $s = \mathbb{E}|X|$ , an interval/Taylor-jet checking engine whose soundness is proved inside Lean, and two families of finite certificates: 569 per-sample-size parameter covers with 18 845 leaves for  $2 \leq n \leq 255$ , and a single certificate of 281 cells that is uniform in  $n$  for all  $n \geq 256$ . The regimes  $n = 1$  and  $\beta/\sqrt{n} \geq 1.3003$  are handled analytically with no computation. There is no `sorry` and no user-declared axiom; the finite certificates are discharged by `native_decide`, so they additionally trust Lean’s compiler. We state the trust boundary precisely and identify which analytic ingredients originate in the earlier work of Xiao and Li.

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# 1 Introduction

## 1.1 The constant

Let  $X, X_1, X_2, \dots$  be independent real random variables with a common law  $\mu$  satisfying

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad \beta := \mathbb{E}|X|^3 < \infty, \quad (1)$$

and set

$$S_n = \frac{X_1 + \dots + X_n}{\sqrt{n}}, \quad F_n(x) = \mathbb{P}(S_n \leq x), \quad \Delta_n = \sup_{x \in \mathbb{R}} |F_n(x) - \Phi(x)|.$$

The Berry–Esseen theorem [1, 3] asserts the existence of a universal constant  $C$  with  $\Delta_n \leq C\beta/\sqrt{n}$  for all  $n \geq 1$  and all laws satisfying (1). We write  $C_{\text{BE}}$  for the least such constant, i.e.

$$C_{\text{BE}} = \sup \left\{ \frac{\sqrt{n}}{\beta} |F_n(x) - \Phi(x)| : \mu \text{ satisfies (1), } n \geq 1, x \in \mathbb{R} \right\} \in [0, \infty]. \quad (2)$$

Esseen [4] proved the lower bound

$$C_{\text{BE}} \geq c_{\text{E}} := \frac{\sqrt{10} + 3}{6\sqrt{2\pi}} = 0.4097321\dots,$$

attained asymptotically along suitably centered and normalized Bernoulli sums, and it is conjectured (usually attributed to Zolotarev [17]) that  $C_{\text{BE}} = c_{\text{E}}$ . In the special case of Bernoulli (equivalently binomial) summands the value  $c_{\text{E}}$  is known to be optimal; see Schulz [7] and Zolotukhin, Nagaev and Chebotarev [18].

Table 1 lists the published upper bounds. The bibliographic data follow the survey entry [12].

| Bound  | Reference             | Comment                                                                                           |
|--------|-----------------------|---------------------------------------------------------------------------------------------------|
| 0.82   | Zolotarev [17]        | Zolotarev-type smoothing inequalities; also 0.9051 for independent (non-i.i.d.) summands.         |
| 0.7975 | van Beek [15]         | Fourier-analytic refinement.                                                                      |
| 0.7655 | Shiganov [11]         |                                                                                                   |
| 0.7056 | Shevtsova [8]         |                                                                                                   |
| 0.5129 | Korolev–Shevtsova [5] | From the structural bound $\Delta_n \leq 0.34445(\beta + 0.489)/\sqrt{n}$ .                       |
| 0.4785 | Tyurin [14]           | From $\Delta_n \leq \min\{0.4690\beta, 0.3322(\beta + 0.429), 0.3031(\beta + 0.646)\}/\sqrt{n}$ . |
| 0.4748 | Shevtsova [10]        |                                                                                                   |
| 0.4690 | Shevtsova [9]         |                                                                                                   |

Table 1: Published upper bounds for  $C_{\text{BE}}$ .

## 1.2 The result

**Theorem 1.1.**  $C_{\text{BE}} \leq \frac{423}{1000}$ .

Theorem 1.1 is formalized in Lean 4 over Mathlib [13]. The constant (2) is defined in the repository as an extended nonnegative real supremum, with finiteness a consequence rather than a hypothesis:

```
def berryEsseenConstant :  $\mathbb{R}_{\geq 0\infty}$  :=
   $\bigsqcup$  ( $\mu$  : ProbabilityMeasure  $\mathbb{R}$ ) ( $\_$  : Admissible  $\mu$ ) ( $n$  :  $\mathbb{N}$ ) ( $\_$  :  $0 < n$ ) ( $x$  :  $\mathbb{R}$ ),
    normalizedError  $\mu$   $n$   $x$ 
```

and the theorem proved is

```
theorem BerryEsseen.Upper0423.berryEsseenConstant_le_0423 :
  BerryEsseen.berryEsseenConstant  $\leq$  0.423
```

in `BerryEsseen/Upper0423/Main.lean`. Section 3 reproduces all the ambient definitions verbatim so that the statement can be audited without reading Lean beyond these few lines.

### 1.3 What is proved and what is computed

The proof has two very different kinds of ingredient, and we keep them rigorously separated throughout.

- (a) *Analysis*. The Prawitz smoothing inequality, a list of pointwise estimates for the characteristic function of an admissible law in terms of  $(\beta, \eta, s)$ , the geometry of the moment region, elementary Gaussian bounds, the semantics of a fixed-point interval arithmetic and of fourth-order Taylor jets, the soundness of a Taylor/range quadrature rule, and the algebra that turns a certified numerical budget into the final inequality. All of this is proved in Lean by ordinary means and is checked by the Lean kernel.
- (b) *Finite computation*. Exactly 579 closed Boolean statements of the form “this explicit certificate is accepted by this explicit checker” are discharged by the tactic `native_decide`, which compiles the Boolean expression to machine code and runs it. Each such use introduces one auxiliary axiom, so the finite part of the proof trusts the Lean compiler and runtime in addition to the kernel.

Nothing about the *search* for the certificates is trusted: the search was done outside Lean, but every accepted numerical bound is recomputed by a checker whose soundness is a Lean theorem, quantified over *all* inputs the checker accepts. Section 13 states the trust boundary precisely and lists what we do and do not claim.

### 1.4 Outline

Section 2 records the provenance of the analytic ingredients. Section 3 fixes notation and reproduces the formal definitions. Section 4 develops the moment geometry ( $\beta \geq 1$ ,  $0 \leq \eta \leq s \leq 1$ ,  $s\beta \geq 1$ ,  $\beta + \eta \geq 2$ ). Section 5 states the smoothing inequality as formalized. Section 6 collects the characteristic-function estimates. Section 7 gives the case decomposition, and Sections 8 and 9 dispose of the two elementary cases. Section 10 describes the interval/jet engine and its soundness, Section 11 the per-sample-size covers for  $2 \leq n \leq 255$ , and Section 12 the uniform certificate for  $n \geq 256$ . Section 13 is the trust boundary and Section 14 the reproduction instructions. Section 15 discusses related and concurrent work.

## 2 Provenance and attribution

Because this is a formalization of a computer-assisted argument that builds on earlier work, it is important to say at the outset which mathematical ingredients are inherited and which are not.

## 2.1 Ingredients originating in Xiao–Li

A substantial part of the analytic framework used here originates in the work of Haonan Xiao and Chunlin Li [16], specifically the repository

<https://github.com/haonan-xiao/iid-berry-esseen>

at commit `f00d781c8c42908bf4aef2739129442a0919fa38`, which proves the bound  $C_{\text{BE}} \leq 879/2000 = 0.4395$  with its own Lean formalization (accessed 13 September 2026). We refer to that work as [XL]. The following ingredients are due to [XL] and are used here:

- (i) **The moment coordinates and the moment region.** The idea of carrying the symmetrized third-moment ratio alongside  $\beta$  and the first absolute moment  $s = \mathbb{E}|X|$ , and of using the resulting region as the parameter domain of the whole argument, is from [XL], `proof-guide.md` §§2–3. Our  $\eta$  is an affine reparametrization of their ratio  $r = \mathbb{E}|X - X'|^3 / (2\mathbb{E}|X|^3)$ , namely  $\eta = \beta(r - 1)$ .
- (ii) **The Prawitz normalization convention.** The particular form of the doubled Prawitz kernel and of the resulting exact upper functional (Section 5) follows [XL], `proof-guide.md` §§5–6.
- (iii) **The stop-loss identity and the excess bounds.** The identity expressing  $\mathbb{E}|X - X'|^3$  through the stop-loss transform, and the resulting inequalities  $\eta \geq 0$  and  $\beta + \eta \geq 2$ , are from [XL], `lean/BerryEsseen/Moments/ThirdMomentRatio.lean` (`symmetrizedThirdAbsoluteMoment_stopLoss_identity`). See Section 4.
- (iv) **The two-point sine circle.** The mechanism by which a centered law is decomposed into centered two-point laws and the imaginary part of the characteristic function is confined to a circle jointly with an energy coordinate is from [XL], `lean/BerryEsseen/CharacteristicFunctions/SineRemainder.lean` (`sine_remainder_circle` together with the two-point energy and moment lemmas). See Proposition 6.10.
- (v) **The radial estimates.** The separate radial bounds for the real and imaginary parts in terms of  $1 - s$  are from [XL], `lean/BerryEsseen/CharacteristicFunctions/TwoPointComparison.lean` (`charFun_real_radial_bound`, `charFun_imag_radial_bound`). See Proposition 6.11.
- (vi) **The disk/strip intersection.** The idea of intersecting the second-order Taylor disk with the real and imaginary strips, which is what the *coordinate* modulus branch of our engine implements, is from [XL], `lean/BerryEsseen/CharacteristicFunctions/MomentDisk.lean`.
- (vii) **The cubic cosine constant and the convex profile.** The constant

$$\kappa = \max_{v>0} \frac{\cos v - 1 + v^2/2}{v^3}$$

and its use as a global majorant appear in [XL], whose proof guide brackets it as  $0.09916191350 < \kappa < 0.09916191353$  and uses the rational upper bound `kappaPlus` =  $9916191353/10^{11}$ . The convex profile obtained by splicing the supporting line  $\frac{1}{2} - \kappa v$  to  $(1 - \cos v)/v^2$  also appears there. Our development uses the coarser rational majorant  $\kappa \leq 49581/500000 = 0.099162$ , which is what the certificates are calibrated to. See Section 6.1.

Our Lean proofs of all these lemmas were written from scratch in the present development and are checked by the Lean kernel; we do not import, translate or otherwise reuse [XL]’s Lean code. What is inherited is mathematical: the choice of coordinates, the smoothing normalization, and the specific mechanisms listed above. We make no claim of priority for any of them.

## 2.2 What is not used

Theorem 1.1 does not use [XL]’s numerical result, nor any of their certificates, nor their region decomposition. This is not a stylistic choice: their certificates are budgets for the target 0.4395 and cannot be reused as budgets for 0.423, and their Lyapunov cutoff  $\beta/\sqrt{n} \leq 56/45$  is tailored to 0.4395 (our analytic cap  $\Delta_n \leq 11/20$  forces the smaller cutoff 13003/10000; see Section 9). We make no assertion about the correctness of their 0.4395 theorem, which we have not audited.

Theorem 1.1 also does *not* use Shevtsova’s published bound [10, 9], or any other result from Table 1. Earlier drafts of the underlying research notes did invoke Shevtsova’s estimate to dispose of the large-third-moment region; the formalized proof replaces that input by the direct treatment of Sections 9 and 11. The only external mathematical library used is Mathlib.

## 2.3 What is new here

Relative to [XL] the following ingredients are, as far as we can tell, new in the present development. We state this descriptively and claim no more.

- (a) **The anchored imaginary envelope.** The cubic imaginary bound  $|\operatorname{Im} f(u)| \leq \frac{1}{6}\sqrt{(\beta-1)(\beta+5/3)}|u|^3$  anchored at  $\cos u$  rather than at 1 (Proposition 6.8), and the sharpened quadratic bound  $|\operatorname{Im} f(u)| \leq \frac{1}{2}\sqrt{2(\beta-1)/3}u^2$  (Proposition 6.9).
- (b) **The signed real interval.** The two-sided estimate  $\cos u \leq \operatorname{Re} f(u)$  on  $[0, \pi]$  (Proposition 6.6) together with the three upper bounds of Proposition 6.7, which lets the engine use the *signed* real part in the coordinate modulus branch.
- (c) **The joint complex anchor disk.** The ODE-based bound  $\|f(u) - \cos u\| \leq \sqrt{(\beta-1)(\beta+5/3)}(u - \sin u)$  on  $[0, \pi]$  (Proposition 6.12).
- (d) **The geometric-sum comparison.** The replacement of the crude  $|a^n - b^n| \leq n|a - b|$  by the exact telescoping sum  $\sum_{j < n} |a|^j |b|^{n-1-j}$ , evaluated by a halving recursion inside the certified jet arithmetic (Section 10.6).
- (e) **The uniform-in- $n$  certificate.** A single finite certificate covering *all*  $n \geq 256$  simultaneously, obtained by reducing the smoothing functional to five one-dimensional integrals in the two scalars  $L = (\beta + \eta)/\sqrt{n}$  and  $h = 1/\sqrt{n}$  (Section 12).
- (f) **The elimination of the published external input**, as described above.

## 3 Notation and the formal statement

### 3.1 The ambient definitions

The following definitions are those of `BerryEsseen/Defs.lean` (identical to the independently auditable copy in `Challenge.lean`), reproduced here verbatim.

```
def thirdAbsMoment (μ : ProbabilityMeasure ℝ) : ℝ := ∫ x, |x| ^ 3 ∂(μ : Measure ℝ)

structure Admissible (μ : ProbabilityMeasure ℝ) : Prop where
```

```

integrable_id : Integrable (fun x : ℝ ↦ x) (μ : Measure ℝ)
integrable_sq : Integrable (fun x : ℝ ↦ x ^ 2) (μ : Measure ℝ)
integrable_abs_cube : Integrable (fun x : ℝ ↦ |x| ^ 3) (μ : Measure ℝ)
mean_zero : (∫ x, x ∂(μ : Measure ℝ)) = 0
second_moment_one : (∫ x, x ^ 2 ∂(μ : Measure ℝ)) = 1

def normalizedSum (n : ℕ) (x : Fin n → ℝ) : ℝ := (∑ i, x i) / Real.sqrt n

def normalizedSumLaw (μ : ProbabilityMeasure ℝ) (n : ℕ) : Measure ℝ :=
  Measure.map (normalizedSum n) (Measure.pi fun _ : Fin n ↦ (μ : Measure ℝ))

def normalCDF (x : ℝ) : ℝ := (gaussianReal 0 1).real (Set.Iic x)

def normalizedError (μ : ProbabilityMeasure ℝ) (n : ℕ) (x : ℝ) : ℝ≥0∞ :=
  ENNReal.ofReal (Real.sqrt n / thirdAbsMoment μ *
    |(normalizedSumLaw μ n).real (Set.Iic x) - normalCDF x|)

def berryEsseenConstant : ℝ≥0∞ :=
  ⌊ (μ : ProbabilityMeasure ℝ) ( _ : Admissible μ) (n : ℕ) ( _ : 0 < n) (x : ℝ),
    normalizedError μ n x

```

Three points deserve emphasis. First, independence is realized by the finite product measure  $\mu^{\otimes n}$  on  $\mathbb{R}^{\text{Fin } n}$ , so no auxiliary probability space is assumed to exist. Second, the supremum is taken in  $[0, \infty]$ , so finiteness of  $C_{\text{BE}}$  is part of the conclusion and not a hypothesis. Third, `normalizedError` is defined with  $\sqrt{n}/\beta$  as a multiplicative prefactor rather than by an inequality  $\Delta_n \leq C\beta/\sqrt{n}$ ; the two readings agree because  $\beta > 0$  for every admissible law (Lemma 4.1).

The reduction from the supremum to a pointwise statement is the trivial

$$C_{\text{BE}} \leq C \iff \forall \mu \text{ admissible}, \forall n \geq 1, \forall x, \quad \text{normalizedError } \mu \ n \ x \leq C, \quad (3)$$

which is `BerryEsseen.constant_le_iff` in `BerryEsseen/Constant.lean`. Everything below therefore concerns the single pointwise quantity

$$\mathcal{E}(\mu, n, x) := \frac{\sqrt{n}}{\beta} |F_n(x) - \Phi(x)|,$$

which we must bound by 423/1000.

### 3.2 Moment coordinates

Throughout,  $\mu$  is an admissible law,  $X \sim \mu$ ,  $X'$  an independent copy, and  $f(u) = \mathbb{E}e^{iuX}$  is the characteristic function. We use the three moment coordinates

$$\beta = \mathbb{E}|X|^3, \quad \eta = \frac{1}{2}\mathbb{E}|X - X'|^3 - \beta, \quad s = \mathbb{E}|X|, \quad (4)$$

and the derived quantity

$$\Gamma = \sqrt{(\beta - 1)^2 + \beta^2 - \eta^2}. \quad (5)$$

We call  $\eta$  the *symmetrization excess*; it is the excess of the half-symmetrized third absolute moment over  $\beta$  itself, and it vanishes for the Rademacher law. Carrying  $(\beta, \eta, s)$  jointly — equivalently, carrying  $\beta$ , the symmetrized third-moment ratio  $r = 1 + \eta/\beta$  and the first absolute moment — is the choice of coordinates made by [XL] (item (i) of Section 2), and the region (6) below is their moment region.

*In Lean:* `thirdAbsMoment`, `symmExcess`, `absMean`, `gammaR` in `BerryEsseen/Defs.lean`, `Upper0423/Analytic/Moments.lean`, `Upper0423/LargeN/Defs.lean`.

We also write

$$\ell = \frac{\beta}{\sqrt{n}}$$

for the Lyapunov fraction, and reserve  $g(u) = e^{-u^2/2}$  for the standard normal characteristic function.

### 3.3 Constants

Table 2 lists all the numerical constants that occur in the proof; each is a rational number written exactly as it appears in the Lean sources.

| Symbol        | Exact value                           | Role                                                                                                       |
|---------------|---------------------------------------|------------------------------------------------------------------------------------------------------------|
| target        | 423/1000                              | the bound being proved ( <code>Native.target</code> )                                                      |
| $\kappa$      | 49581/500000                          | rational majorant for $\max_v(\cos v - 1 + v^2/2)/v^3$ ( <code>Native.kappa</code> , <code>kappaR</code> ) |
| $B$           | 712569/454000                         | upper end of the original small- $\beta$ covers ( <code>deltaMax</code> = $B - 1 = 258569/454000$ )        |
| $\varepsilon$ | 1/1000                                | low-frequency omission cutoff ( <code>Native.epsilon</code> )                                              |
| —             | 13003/10000                           | Lyapunov threshold for the Cantelli case                                                                   |
| —             | 11/20                                 | the universal Cantelli cap for $\Delta_n$                                                                  |
| —             | 21/50                                 | the $n = 1$ constant                                                                                       |
| $A$           | $(8 - 64\kappa)/(1 - \cos 4)$         | scale of the curved branch of the convex profile                                                           |
| $2^{40}$      | 1099511627776                         | fixed-point scale of the interval arithmetic ( <code>Native.scale</code> )                                 |
| $\tau$        | 1/4                                   | smoothing split in the uniform certificate                                                                 |
| $r$           | 16                                    | $r^2 = 256$ , the first sample size of the uniform regime                                                  |
| $m$           | 256                                   | panel count per cell in the uniform certificate                                                            |
| $\nu$         | $(\frac{22/7}{3(1-(22/7)^2/96)})^2/2$ | quadratic coefficient of the normal-correction majorant ( <code>LargeN.nu</code> )                         |

Table 2: The numerical constants, exactly as in the Lean sources.

## 4 The moment region

All the pointwise estimates of Section 6 are stated in terms of  $(\beta, \eta, s)$ , so the first task is to delimit the region in which these parameters live. The following four facts are what the certificates use.

**Lemma 4.1** ( $\beta \geq 1$ ). *For every admissible  $\mu$  we have  $\beta \geq 1$ , and in particular  $\beta > 0$ .*

*Proof.* The polynomial  $w(y) = (y - 1)^2(2y + 1) = 2y^3 - 3y^2 + 1$  is nonnegative for  $y \geq 0$ . Applying it to  $y = |X|$  and using  $\mathbb{E}X^2 = 1$  gives  $0 \leq \mathbb{E}w(|X|) = 2\beta - 3 + 1 = 2(\beta - 1)$ .  $\square$

*In Lean:* `momentExcessWeight`, `one_le_thirdAbsMoment`, `thirdAbsMoment_pos` in `BerryEsseen/MomentExcess.lean`.

**Lemma 4.2** ( $0 \leq \eta \leq s \leq 1$  and  $s\beta \geq 1$ ). *For every admissible  $\mu$ ,*

$$0 \leq \eta \leq s \leq 1, \quad s\beta \geq 1.$$

*Proof.* For  $s \leq 1$  apply Cauchy–Schwarz:  $s = \mathbb{E}|X| \leq (\mathbb{E}X^2)^{1/2} = 1$ . For  $s\beta \geq 1$  apply Cauchy–Schwarz in the form  $1 = (\mathbb{E}X^2)^2 = (\mathbb{E}|X|^{1/2} \cdot |X|^{3/2})^2 \leq \mathbb{E}|X| \cdot \mathbb{E}|X|^3 = s\beta$ . For  $\eta \geq 0$  and  $\eta \leq s$  we use two elementary scalar inequalities:

$$|x - y|^3 \geq |x|^3 + |y|^3 - 3xy(|x| + |y|), \quad |x - y|^3 \leq (x - y)^2(|x| + |y|).$$

Integrating the first against  $\mu \otimes \mu$  and using  $\mathbb{E}X = 0$  gives  $\mathbb{E}|X - X'|^3 \geq 2\beta$ , i.e.  $\eta \geq 0$ ; integrating the second and expanding  $(x - y)^2$  gives  $\mathbb{E}|X - X'|^3 \leq 2\beta + 2s$ , i.e.  $\eta \leq s$ .  $\square$

*In Lean:* `abs_sub_cube_ge`, `abs_sub_cube_le`, `symmExcess_nonneg`, `symmExcess_le_absMean`, `symmExcess_le_one`, `one_le_absMean_mul_thirdAbsMoment` in `Upper0423/Analytic/Moments.lean`.

The excess bounds of Lemma 4.2, in the equivalent formulation through the ratio  $r = \mathbb{E}|X - X'|^3/(2\beta)$ , are due to [XL] (item (iii) of Section 2); the proof above is an independent and shorter rederivation of the two inequalities actually needed.

**Proposition 4.3** (Feasibility:  $\beta + \eta \geq 2$ ). *For every admissible  $\mu$  we have  $\mathbb{E}|X - X'|^3 \geq 4$ , equivalently  $\beta + \eta \geq 2$ , with equality for the Rademacher law.*

*Proof sketch.* For  $x > y$  one has  $\frac{1}{6}(x - y)^3 = \int_y^x (x - t)(t - y) dt$ , so

$$|x - y|^3 = 6 \int_{\mathbb{R}} [(x - t)_+(t - y)_+ + (y - t)_+(t - x)_+] dt,$$

and taking expectations,  $\mathbb{E}|X - X'|^3 = 12 \int_{\mathbb{R}} P(t)N(t) dt$  with  $P(t) = \mathbb{E}(X - t)_+$  and  $N(t) = \mathbb{E}(t - X)_+ = P(t) + t$ . Subtracting the Rademacher stop-loss profile  $a_0(t) = \frac{1}{2}(1 - t)_+$  turns the deficit into a Gram form: with

$$G(x, y) = \int_0^\infty ((x - t)_+ - a_0(t))((y - t)_+ - a_0(t)) dt,$$

one has the pointwise minorant  $|x - y|^3 \geq (\text{explicit}) + 12 G(x, y)$ , and  $\iint G d\mu d\mu \geq 0$  by Fubini because  $G$  is a positive semidefinite kernel (a square of an integral of a single function). The explicit part integrates to exactly 4.  $\square$

*In Lean:* `radStopLoss`, `stopLossGram`, `stopLossGram_integral_nonneg`, `four_le_integral_symm_abs_cube`, `two_le_thirdAbsMoment_add_symmExcess` in `Upper0423/Analytic/StopLoss.lean`.

**Remark 4.4.** The stop-loss formulation of Proposition 4.3 is due to [XL] (item (iii) of Section 2); the proof above is a self-contained rederivation. Proposition 4.3 is what makes the coordinate  $a = \beta + \eta$  useful: it is bounded below by 2 and it is precisely the quantity that controls the modulus of  $f$  through the convex profile (Section 6.1).

Summarizing, the parameter region used by the certificates is

$$\beta \geq 1, \quad 0 \leq \eta \leq s \leq 1, \quad s\beta \geq 1, \quad \beta + \eta \geq 2. \quad (6)$$

Two coordinate systems on (6) are used:

- for  $\beta \leq 2$ :  $\delta = \beta - 1 \in [0, 1]$  and  $w \in [0, 1]$  with  $\eta = 1 - \delta w$  (so  $w$  interpolates between  $\eta = 1$  and  $\eta = 1 - \delta$ ; note  $\eta \geq 2 - \beta = 1 - \delta$  by (6));
- for  $\beta \geq 2$ : the pair  $(\beta, \eta)$  directly, with  $\eta \in [0, 1]$ .

## 5 The Prawitz smoothing inequality

### 5.1 The kernel

**Definition 5.1.** For  $s \in (0, 1)$  put

$$k(s) = (1 - s) \frac{\cos \pi s}{\sin \pi s} + \frac{1}{\pi}, \quad K_\sigma(s) = \sigma(1 - s) + i k(s) \quad (\sigma \in \{\pm 1\}).$$

We abbreviate  $K = K_1$ .

*In Lean:* `prawitzSineKernel`, `prawitzKernel` in `BerryEsseen/PrawitzKernel.lean`,  
`BerryEsseen/PrawitzFourier.lean`.

Since  $|K_{-1}(s)| = |K_1(s)|$  we only ever need

$$|K(s)| = \sqrt{(1 - s)^2 + k(s)^2}. \quad (7)$$

Two equivalent rewritings of (7), both used by the engine, follow from the cotangent defect  $q(z) = \frac{1}{z} - \cot z$ :

$$|K(s)| = (1 - s) \sqrt{1 + q(\pi(1 - s))^2}, \quad (8)$$

$$s |K(s)| = \sqrt{(s(1 - s))^2 + \left(\frac{1}{\pi} - s(1 - s) q(\pi s)\right)^2}. \quad (9)$$

Indeed  $k(s) = (1 - s)q(\pi(1 - s))$  using  $\cot(\pi - z) = -\cot z$ , which gives (8); and  $k(s) = \frac{1}{\pi s} - (1 - s)q(\pi s)$ , which gives (9) after multiplying by  $s$ . Form (9) is numerically stable near  $s = 0$  (where  $|K|$  blows up like  $1/(\pi s)$  but  $s|K(s)|$  is smooth) and form (8) near  $s = 1$ . This is why the high-frequency panels of a certificate are never allowed to straddle  $s = 1/2$ : on  $[\tau, 1/2]$  the engine uses (9), on  $[1/2, 1]$  it uses (8).

Finally, the normal correction uses

$$\left| K(s) - \frac{i}{\pi s} \right| = (1 - s) \sqrt{1 + q(\pi s)^2}, \quad (10)$$

because  $\frac{1}{\pi} - \frac{1}{\pi s} = -\frac{1-s}{\pi s}$ , so the imaginary part of  $K(s) - i/(\pi s)$  equals  $-(1 - s)q(\pi s)$ .

*In Lean:* `kernel_norm_eq_sqrt`, `kernelR_left`, `kernelR_right`, `kernelR_scaled`,  
`kernelR_normal` in `Upper0423/LargeN/Kernels.lean`, `Upper0423/Analytic/Kernels.lean`.

### 5.2 The smoothing bound

**Definition 5.2.** For a probability measure  $\rho$  on  $\mathbb{R}$  with characteristic function  $\varphi$ , and for  $T > 0$ ,  $\tau \in (0, 1]$ , put

$$\begin{aligned} \mathcal{S}(\rho, T, \tau) &= \int_0^\tau |K(t)| |\varphi(Tt) - e^{-T^2 t^2/2}| dt && \text{(low)} \\ &+ \int_\tau^1 |K(t)| |\varphi(Tt)| dt && \text{(high)} \\ &+ \int_0^\tau \left| K(t) - \frac{i}{\pi t} \right| e^{-T^2 t^2/2} dt && \text{(normal correction)} \\ &+ \int_\tau^\infty \frac{e^{-T^2 t^2/2}}{\pi t} dt. && \text{(tail)} \end{aligned}$$

*In Lean:* `smoothingIntegral`, `prawitzLowError`, `prawitzHighMagnitude`,  
`prawitzNormalCorrection`, `prawitzNormalTail` in `BerryEsseen/PrawitzSmoothing.lean`,  
`BerryEsseen/SmoothingIntegrability.lean`.

**Theorem 5.3** (Prawitz smoothing). *Let  $\rho$  be a probability measure on  $\mathbb{R}$  with  $\int |y| d\rho < \infty$ , let  $R$  be its distribution function,  $T > 0$  and  $\tau \in (0, 1]$ . Then for every  $x \in \mathbb{R}$ ,*

$$|R(x) - \Phi(x)| \leq \mathcal{S}(\rho, T, \tau).$$

*In Lean: `prawitz_smoothing`, `abs_prawitzFourier_residual_le`, `normalizedSum_prawitz_smoothing` in `BerryEsseen/PrawitzSmoothing.lean`, `BerryEsseen/SmoothingMajorants.lean`.*

Theorem 5.3 is the inequality of Prawitz [6], in the normalization of [XL] (item (ii) of Section 2). The formalization proceeds through the Fourier functional

$$\mathcal{P}(\rho, \sigma, T, x) = \int_0^1 \operatorname{Re} \left[ K_\sigma(t) e^{-iTtx} \varphi(Tt) \right] dt,$$

which satisfies  $R(x) - \frac{1}{2} \leq \mathcal{P}(\rho, 1, T, x)$  and  $\mathcal{P}(\rho, -1, T, x) \leq R(x) - \frac{1}{2}$  by Prawitz's two-sided inversion formula (formalized through a Fubini argument and an explicit majorant/minorant of the indicator). Comparing both sides with the Gaussian value  $\Phi(x) - \frac{1}{2}$ , replacing  $\varphi$  by  $e^{-T^2 t^2/2}$  in the range  $t \leq \tau$  and by 0 in the range  $t > \tau$ , and using  $|K_{\pm 1}| = |K|$ , produces the four terms of Definition 5.2. The third and fourth terms are the price of the substitution: the exact Gaussian Fourier functional is recovered up to  $|K(t) - i/(\pi t)|$  on  $[0, \tau]$  and the pure Gaussian tail beyond  $\tau$ , the latter because  $\int_0^\infty \frac{e^{-T^2 t^2/2}}{\pi t} \sin(Ttx) dt$  is the exact inversion integral for  $\Phi$ .

### 5.3 The i.i.d. scaling

For an admissible  $\mu$  and  $n \geq 1$ , write  $\rho_n$  for the law of  $S_n$ . Then  $\varphi_n(v) = f(v/\sqrt{n})^n$ , so with the choice

$$T = c\sqrt{n} \quad (c > 0) \tag{11}$$

we get  $\varphi_n(Tt) = f(ct)^n$  and  $e^{-T^2 t^2/2} = g(ct)^n$  with  $g(u) = e^{-u^2/2}$ . Thus every integrand in Definition 5.2 becomes a function of the *per-summand* frequency

$$u = ct, \tag{12}$$

raised to the power  $n$ , times a kernel factor depending on  $t$  alone:

$$\begin{aligned} \text{low}(t) &= |K(t)| \cdot |f(u)^n - g(u)^n|, \\ \text{high}(t) &= |K(t)| \cdot |f(u)|^n, \\ \text{normal}(t) &= \left| K(t) - \frac{i}{\pi t} \right| \cdot g(u)^n. \end{aligned} \tag{13}$$

*In Lean: `charFun_normalizedSum_scaled`, `prawitzLowError_scaled`, `prawitzHighMagnitude_scaled` in `Upper0423/Small/Leaf.lean`.*

The tail term has a closed reduction: substituting  $v = T^2 t^2/2$ ,

$$\int_\tau^\infty \frac{e^{-T^2 t^2/2}}{\pi t} dt = \frac{1}{2\pi} \int_{nc^2\tau^2/2}^\infty \frac{e^{-v}}{v} dv = \frac{E_1(nc^2\tau^2/2)}{2\pi}. \tag{14}$$

*In Lean: `prawitzNormalTail_integral_eq` in `Upper0423/Small/Tail.lean`.*

## 6 Characteristic-function estimates

This section collects the pointwise estimates for  $f$  that the certificates consume. In Lean they are bundled into a single structure

```
structure ChfFacts (f : ℝ → ℂ) (β η s : ℝ) : Prop
```

with eighteen fields, and the theorem `chfFacts` proves that every admissible law satisfies it with the parameters (4). The engine's soundness is proved from `ChfFacts` alone, so this list is exactly the analytic interface between the mathematics and the computation.

*In Lean:* `ChfFacts` in `Upper0423/Analytic/ModulusBridge.lean`.

*In Lean:* `chfFacts` in `Upper0423/Analytic/ChfFactsProof.lean`.

### 6.1 The cubic cosine constant and the convex profile

**Lemma 6.1.** *For all  $v \in \mathbb{R}$ ,  $\cos v - 1 + \frac{v^2}{2} \leq \kappa|v|^3$  with  $\kappa = 49581/500000$ .*

*Proof sketch.* By symmetry assume  $v \geq 0$ . For  $v \leq 2$  the Taylor bound  $\cos v \leq 1 - \frac{v^2}{2} + \frac{v^4}{24}$  suffices since  $v^4/24 \leq \kappa v^3$  for  $v \leq 24\kappa$ , and  $24\kappa > 2$ . For  $v \geq 6$  the left side is at most  $\frac{v^2}{2}$  and  $\frac{1}{2} \leq \kappa v$ . The interval  $[2, 6]$  is covered by seven polynomial certificates, each reducing to the nonnegativity of an explicit polynomial on a dyadic subinterval after substituting the degree-20 Taylor majorant of  $\cos$ .  $\square$

*In Lean:* `cos_le_taylor_twenty`, `kappa_gap_nonneg_0 ... kappa_gap_nonneg_6`, `cos_sub_one_add_half_sq_le_kappa` in `Upper0423/Analytic/CosineProfile.lean`.

The constant  $\kappa$  and this use of it are due to [XL] (item (vii) of Section 2); the true maximum is  $0.09916191350\dots$ , and  $49581/500000 = 0.099162$  is a slightly coarser rational majorant which is what all certificates are calibrated against.

Averaging Lemma 6.1 over  $v = uX$  gives at once:

**Proposition 6.2.** *For all  $u \in \mathbb{R}$ ,  $\operatorname{Re} f(u) \leq 1 - \frac{u^2}{2} + \kappa\beta|u|^3$ , and  $|f(u)|^2 \leq 1 - u^2 + 2\kappa(\beta + \eta)|u|^3$ .*

*Proof.* The first bound is Lemma 6.1 with  $v = uX$ , using  $\mathbb{E}X^2 = 1$ . For the second, note  $|f(u)|^2 = \mathbb{E} \cos(u(X - X'))$  for an independent copy  $X'$ ; apply Lemma 6.1 with  $v = u(X - X')$  and use  $\mathbb{E}(X - X')^2 = 2$  and  $\mathbb{E}|X - X'|^3 = 2(\beta + \eta)$ .  $\square$

*In Lean:* `charFun_re_le_kappa`, `normSq_charFun_le_kappa` in `Upper0423/Analytic/CosineProfile.lean`.

The second bound of Proposition 6.2 degenerates for large  $|u|$ . The fix, again due to [XL], is to replace the supporting line  $\frac{1}{2} - \kappa v$  by a globally convex profile.

**Definition 6.3** (Convex profile). With  $A = (8 - 64\kappa)/(1 - \cos 4)$  put

$$q(v) = \begin{cases} \frac{1}{2} - \kappa v, & 0 \leq v \leq 4, \\ A \frac{1 - \cos v}{v^2}, & 4 < v \leq 2\pi, \\ 0, & v > 2\pi, \end{cases} \quad q(v) = q(|v|) \text{ for } v < 0.$$

**Lemma 6.4.**  *$q$  is continuous, convex on  $\mathbb{R}$ , nonincreasing and nonnegative on  $[0, \infty)$ , and satisfies  $\cos v \leq 1 - v^2 q(|v|)$  for all  $v$ .*

*Proof sketch.* The value  $A$  is chosen so that the two branches agree at  $v = 4$ : indeed  $A(1 - \cos 4) = 8 - 64\kappa$  gives  $A(1 - \cos 4)/16 = \frac{1}{2} - 4\kappa$ . One checks  $\frac{d}{dv}(A(1 - \cos v)/v^2) \leq 0$  and  $\frac{d^2}{dv^2}(A(1 - \cos v)/v^2) \geq 0$  on  $[4, 2\pi]$  by explicit sign estimates for  $\sin$  and  $\cos$  there, and that the right derivative at 4 dominates  $-\kappa$ ; convexity of the splice follows, and the third branch is joined because  $A(1 - \cos 2\pi)/(2\pi)^2 = 0$  with vanishing derivative. The inequality  $\cos v \leq 1 - v^2 q(|v|)$  holds on  $[0, 4]$  by Lemma 6.1, is an equality on  $(4, 2\pi]$  by definition of  $A$  (up to the factor  $A \leq 1$ ), and is trivial beyond  $2\pi$ .  $\square$

*In Lean:* `profile, convex0n_profile, antitone_profile, cos_le_profile` in `Upper0423/Analytic/CosineProfile.lean`.

**Proposition 6.5** (Global modulus bound). *For all  $u$ ,*

$$|f(u)|^2 \leq 1 - 2u^2 q((\beta + \eta)|u|).$$

*In particular, if  $\beta + \eta \leq D$  and  $4 \leq D|u| \leq 2\pi$ , then*

$$|f(u)|^2 \leq 1 - \frac{2A(1 - \cos(D|u|))}{D^2}.$$

*Proof sketch.* Write  $|f(u)|^2 = \mathbb{E} \cos(u(X - X')) \leq 1 - u^2 \mathbb{E}[(X - X')^2 q(|u(X - X')|)]$  by Lemma 6.4. The measure  $\frac{1}{2}(X - X')^2 d\mathbb{P}$  has total mass 1 and, by convexity of  $q$  and Jensen's inequality against that weighted measure,  $\mathbb{E}[\frac{1}{2}(X - X')^2 q(|u(X - X')|)] \geq q(|u| \mathbb{E}[\frac{1}{2}(X - X')^2 |X - X'|]) = q(|u|(\beta + \eta))$ , using  $\mathbb{E}|X - X'|^3 = 2(\beta + \eta)$ . The second display uses that  $q$  is nonincreasing, so  $q(D|u|) \leq q((\beta + \eta)|u|)$  is false in the wrong direction — one applies monotonicity in the correct direction, replacing  $(\beta + \eta)|u|$  by the larger  $D|u|$  and using the antitonicity of  $q$  on  $[0, \infty)$  together with the explicit curved branch.  $\square$

*In Lean:* `normSq_charFun_le_profile, normSq_charFun_le_curved` in `Upper0423/Analytic/CosineProfile.lean`, `Upper0423/Analytic/ChfFactsProof.lean`.

## 6.2 The real part

**Proposition 6.6** (Signed real lower bound). *For  $0 \leq u \leq \pi$ ,  $\cos u \leq \operatorname{Re} f(u)$ .*

*Proof.* Fix  $u \in (0, \pi]$  and put  $c = \sin u/(2u)$ . We claim  $H(v) = \cos v + cv^2$  satisfies  $H(v) \geq H(u)$  for all  $v \geq 0$ . The chord bound  $w \sin u \leq u \sin w$  for  $0 \leq w \leq u \leq \pi$  (concavity of  $\sin$  on  $[0, \pi]$ ) gives  $H'(v) = 2cv - \sin v \leq 0$  for  $v \leq u$  and  $\geq 0$  for  $v \geq u$  in the relevant range; the case  $v > \pi$  is handled directly since  $\sin v \leq 1$  and  $2cv \geq 2cu = \sin u$ . Substituting  $v = u|X|$  and taking expectations, using  $\mathbb{E}X^2 = 1$  and  $\operatorname{Re} f(u) = \mathbb{E} \cos(uX) = \mathbb{E} \cos(u|X|)$ , yields  $\operatorname{Re} f(u) + c \geq \cos u + cu^2 \cdot 1$  — i.e.  $\operatorname{Re} f(u) \geq \cos u$ .  $\square$

*In Lean:* `mul_sin_le_mul_sin, cos_add_quad_le, cos_le_charFun_re` in `Upper0423/Analytic/RealLower.lean`.

No bounded-support hypothesis is needed. This estimate is what allows the engine to bracket  $\operatorname{Re} f$  from both sides and hence to use the *signed* real part, rather than only its modulus, in the coordinate modulus branch (Section 10.5).

**Proposition 6.7** (Real upper bounds). *For  $u \geq 0$  we have the three bounds*

$$\operatorname{Re} f(u) \leq \cos u + \frac{(\beta - 1)u^3}{6}, \tag{15}$$

$$\operatorname{Re} f(u) \leq \cos u + (1 - s)(u \sin u + u^2), \tag{16}$$

$$\operatorname{Re} f(u) \leq 1 - \frac{u^2}{2} + \kappa \beta u^3. \tag{17}$$

Bound (15) is the two-sided anchor  $|\operatorname{Re} f(u) - \cos u| \leq (\beta - 1)|u|^3/6$ , which follows from  $|\cos(uy) - \cos u - \frac{1}{2}(1 - y^2)(\dots)|$ -type Taylor estimates weighted by  $\mathbb{E}[|X|^3 - \dots]$ ; (16) is the radial bound of Proposition 6.11; (17) is Proposition 6.2.

### 6.3 The imaginary part

Four bounds are used; all four are needed, because they are sharp in different corners of the region (6).

**Proposition 6.8** (Anchored cubic). *For all  $u$ ,*

$$|\operatorname{Im} f(u)| \leq \frac{\sqrt{(\beta - 1)(\beta + 5/3)}}{6} |u|^3.$$

*Proof sketch.*  $\operatorname{Im} f(u) = \mathbb{E} \sin(uX)$  and  $\mathbb{E}(uX) = 0$ , so  $\operatorname{Im} f(u) = \mathbb{E}[\sin(uX) - uX]$  and  $|\sin v - v| \leq |v|^3/6$  gives the crude  $\beta|u|^3/6$ . The improvement comes from noticing that the relevant moment is  $\mathbb{E}[|X|(X^2 - 1)]$  rather than  $\mathbb{E}|X|^3$ : by Cauchy–Schwarz with the weight  $w(y) = (y - 1)^2(2y + 1)$  of Lemma 4.1,

$$(\mathbb{E}[|X|(X^2 - 1)])^2 \leq \mathbb{E}w(|X|) \cdot \mathbb{E} \frac{\dots}{\dots} \leq (\beta - 1)(\beta + 5/3),$$

the second inequality being an explicit polynomial comparison. The anchoring at  $\cos u$  (rather than at 1) is what makes the moment  $(\beta - 1)$ -small.  $\square$

*In Lean:* `charFun_im_cubic_anchor` in `BerryEsseen/ImaginaryPart.lean` (via `Upper0423/Analytic/ChfFactsProof.lean`).

**Proposition 6.9** (Sharpened quadratic). *For all  $u$ ,  $|\operatorname{Im} f(u)| \leq \frac{\sqrt{2(\beta - 1)/3}}{2} u^2$ .*

*Proof.* The scalar inequality  $|\sin(uy) - y \sin u| \leq \frac{u^2}{2} y|y - 1|$  for  $y \geq 0$ , applied with  $y = |X|$  and combined with  $\mathbb{E}X = 0$ , gives  $|\operatorname{Im} f(u)| \leq \frac{u^2}{2} \mathbb{E}[|X| \cdot ||X| - 1|]$ . The polynomial identity

$$w(y) \frac{y^2 + 2y}{9} - y^2(y - 1)^2 = \frac{2y(y - 1)^4}{9} \geq 0 \quad (y \geq 0),$$

with  $w(y) = (y - 1)^2(2y + 1)$  and  $\mathbb{E}w(|X|) = 2(\beta - 1)$ , gives by weighted Cauchy–Schwarz

$$\left( \mathbb{E}[|X| \cdot ||X| - 1|] \right)^2 \leq 2(\beta - 1) \frac{1 + 2s}{9} \leq \frac{2(\beta - 1)}{3},$$

using  $s \leq 1$ .  $\square$

*In Lean:* `weighted_linear_sq_le_sharp`, `integral_abs_weighted_linear_bound_sharp`, `abs_charFun_im_le_quadratic_sharp` in `Upper0423/Analytic/QuadraticImag.lean`.

**Proposition 6.10** (Sine circle). *For all  $u$ ,*

$$|\operatorname{Im} f(u)| \leq \frac{\sqrt{\beta^2 - \eta^2}}{6} |u|^3.$$

*Proof sketch.* This is the two-point decomposition of [XL] (item (iv) of Section 2). Write  $r_0 = \mathbb{E}X_+ = s/2$  and decompose  $\mu$  as a mixture of centered two-point laws  $P_{a,b}$  (mass  $b/(a + b)$  at  $a$ , mass  $a/(a + b)$  at  $-b$ ) with mixing measure  $d\nu = \frac{a+b}{r_0} d\mu_+(a) d\mu_-(b)$ . For each component, with  $h_{a,b}(u) = \operatorname{Im} \widehat{P}_{a,b}(u)$ ,

$$|h_{a,b}(u)| \leq \frac{|u|^3 ab |a^2 - b^2|}{6(a + b)},$$

and with  $\beta_{a,b} = \frac{ab(a^2+b^2)}{a+b}$  and  $e_{a,b} = \frac{2a^2b^2}{a+b}$  one has the pointwise *circle inequality*

$$e_{a,b}^2 + \left( \frac{6 h_{a,b}(u)}{|u|^3} \right)^2 \leq \beta_{a,b}^2.$$

Minkowski's inequality in  $L^2(\nu)$  upgrades this to  $\zeta^2 + (6 \operatorname{Im} f(u)/|u|^3)^2 \leq \beta^2$  where  $\zeta = \int e_{a,b} d\nu$ , and an explicit computation gives  $\zeta = 4A_2B_2/s \geq s \geq \eta$  (with  $A_2 = \mathbb{E}X_+^2$ ,  $B_2 = \mathbb{E}X_-^2$ ), using Lemma 4.2. Hence  $(6 \operatorname{Im} f(u)/|u|^3)^2 \leq \beta^2 - \eta^2$ .  $\square$

*In Lean:* `abs_sin_twoPoint_le, twoPoint_circle, sineCircle_prod, abs_charFun_im_le_sineCircle` in `Upper0423/Analytic/SineCircle.lean`.

**Proposition 6.11** (Radial disk). *Put  $d = 1 - s$  and  $Y = |X|$ ,  $A = \operatorname{sgn} X$  (so  $X = AY$ ),  $a = \mathbb{E}A$ . For  $0 \leq u \leq \pi$ ,*

$$\begin{aligned} |f(u) - [\cos u + d u \sin u + i a(\sin u - u \cos u)]| &\leq d u^2, \\ \operatorname{Re} f(u) &\leq \cos u + d(u \sin u + u^2), \\ |\operatorname{Im} f(u)| &\leq \sqrt{2d}(\sin u - u \cos u) + d u^2. \end{aligned}$$

*Proof sketch.* The second-order remainder bound  $|e^{iuY} - e^{iu}(1 + iu(Y-1))| \leq \frac{u^2}{2}(Y-1)^2$  and  $\mathbb{E}(Y-1)^2 = 2 - 2s = 2d$  give the first display after averaging, with the  $ia$  coefficient arising from  $\mathbb{E}[A(\dots)]$  and  $\mathbb{E}[AY] = \mathbb{E}X = 0$ . Cauchy–Schwarz applied to  $0 = \mathbb{E}[AY]$  gives  $a^2 + s^2 \leq 1$ , hence  $a^2 \leq 1 - s^2 \leq 2d$ , which yields the third display; the second follows from the real part of the first together with  $\sin u - u \cos u \geq 0$  on  $[0, \pi]$ .  $\square$

*In Lean:* `abs_integral_radialSign_le, charFun_re_le_radial, abs_charFun_im_le_radial` in `Upper0423/Analytic/Radial.lean`.

Proposition 6.11 is the estimate of [XL] (item (v) of Section 2); the form above retains the complex remainder norm and the sharpened sign-mean estimate  $a^2 \leq 1 - s^2$ .

## 6.4 The joint anchor disk

**Proposition 6.12** (ODE anchor). *For  $0 \leq u \leq \pi$ ,*

$$\|f(u) - \cos u\| \leq \sqrt{(\beta - 1)(\beta + 5/3)} (u - \sin u).$$

*Proof sketch.* For fixed real  $x$  the function

$$G(u) = e^{iux} - \cos u - ix \sin u - (1 - x^2)(1 - \cos u)$$

satisfies  $G(0) = G'(0) = 0$  and  $G'' + G = (1 - x^2)(e^{iux} - 1)$ , whence  $G(u) = \int_0^u \sin(u-v)(1 - x^2)(e^{ivx} - 1) dv$ . Using  $|e^{ivx} - 1| \leq v|x|$  and  $\int_0^u v \sin(u-v) dv = u - \sin u$ ,

$$|G(u)| \leq |x| |x^2 - 1| (u - \sin u).$$

Averaging over  $x = X$  kills the terms  $ix \sin u$  and  $(1 - x^2)(1 - \cos u)$  because  $\mathbb{E}X = 0$  and  $\mathbb{E}X^2 = 1$ , leaving  $\|f(u) - \cos u\| \leq \mathbb{E}[|X| |X^2 - 1|] (u - \sin u)$ , and the moment is bounded by  $\sqrt{(\beta - 1)(\beta + 5/3)}$  as in Proposition 6.8.  $\square$

*In Lean:* `odeAnchorRemainder, norm_odeAnchorRemainder_le, norm_charFun_sub_cos_le_ode` in `Upper0423/Analytic/OdeAnchor.lean`.

Finally the plain Taylor disk is retained:

$$\left\| f(u) - 1 + \frac{u^2}{2} \right\| \leq \frac{\beta |u|^3}{6}. \quad (18)$$

## 6.5 Summary of the interface

Table 3 lists the eighteen fields of **ChfFacts**. Each is proved for admissible laws in the file indicated.

| Field                                               | Content                                    | Source file                              |
|-----------------------------------------------------|--------------------------------------------|------------------------------------------|
| <code>one_le_beta</code>                            | $1 \leq \beta$                             | <code>MomentExcess.lean</code>           |
| <code>eta_nonneg, eta_le_s, s_le_one, s_beta</code> | $0 \leq \eta \leq s \leq 1, s\beta \geq 1$ | <code>Analytic/Moments.lean</code>       |
| <code>norm_le_one</code>                            | $ f  \leq 1$                               | <code>Mathlib</code>                     |
| <code>re_ge_cos</code>                              | Prop. 6.6                                  | <code>Analytic/RealLower.lean</code>     |
| <code>re_anchor</code>                              | (15)                                       | <code>RealPart.lean</code>               |
| <code>re_radial</code>                              | (16)                                       | <code>Analytic/Radial.lean</code>        |
| <code>re_kappa</code>                               | (17)                                       | <code>Analytic/CosineProfile.lean</code> |
| <code>im_cubic</code>                               | Prop. 6.8                                  | <code>ImaginaryPart.lean</code>          |
| <code>im_quad</code>                                | Prop. 6.9                                  | <code>Analytic/QuadraticImag.lean</code> |
| <code>im_circle</code>                              | Prop. 6.10                                 | <code>Analytic/SineCircle.lean</code>    |
| <code>im_radial</code>                              | Prop. 6.11                                 | <code>Analytic/Radial.lean</code>        |
| <code>normSq_kappa</code>                           | Prop. 6.2                                  | <code>Analytic/CosineProfile.lean</code> |
| <code>normSq_curved</code>                          | Prop. 6.5                                  | <code>Analytic/ChfFactsProof.lean</code> |
| <code>ode</code>                                    | Prop. 6.12                                 | <code>Analytic/OdeAnchor.lean</code>     |
| <code>quadratic</code>                              | (18)                                       | <code>CharacteristicBounds.lean</code>   |

Table 3: The analytic interface **ChfFacts**.

## 7 The case decomposition

The proof of Theorem 1.1 splits the parameter space  $\{(\mu, n, x)\}$  into four regions, as follows (`normalizedError_le_0423` in `Upper0423/Main.lean`):

| Region                                          | Lean theorem                             | File                               | Method               |
|-------------------------------------------------|------------------------------------------|------------------------------------|----------------------|
| $n = 1$                                         | <code>normalizedError_one_le</code>      | <code>SingleSummand.lean</code>    | polynomial majorants |
| $\ell \geq \frac{13003}{10000}$                 | <code>normalizedError_large_ell</code>   | <code>LargeEll.lean</code>         | Cantelli             |
| $2 \leq n \leq 255, \ell < \frac{13003}{10000}$ | <code>normalizedError_moderate_le</code> | <code>Covers/All.lean</code>       | 569 covers           |
| $n \geq 256, \ell \leq \frac{13003}{10000}$     | <code>normalizedError_uniform</code>     | <code>Uniform/Data/All.lean</code> | 281 cells            |

Here  $\ell = \beta/\sqrt{n}$ . The four cases are exhaustive: if  $n = 1$  the first applies; otherwise either  $\ell \geq 13003/10000$  (second case) or  $\ell < 13003/10000$ , and then the third or fourth applies according to  $n \leq 255$  or  $n \geq 256$ . The overlap of the second and fourth cases at  $\ell = 13003/10000$  is harmless (both are stated with the appropriate non-strict or strict inequality).

The threshold  $13003/10000$  is exactly what the Cantelli cap allows; see Section 9. Its role in the finite covers is that for  $2 \leq n \leq 255$  one needs only

$$1 \leq \beta \leq \frac{13003}{10000}\sqrt{n}, \quad 0 \leq \eta \leq 1,$$

a compact rational box for each  $n$ .

## 8 The single summand: $n = 1$

**Theorem 8.1.** *For every admissible  $\mu$  and every  $t \in \mathbb{R}$ ,*

$$|\mathbb{P}(X \leq t) - \Phi(t)| \leq \frac{21}{50}\beta.$$

Consequently  $\mathcal{E}(\mu, 1, x) \leq 21/50 \leq 423/1000$ .

Since  $21/50 = 0.42 < 0.423$ , this case needs no computation at all. The mechanism is a family of *rational polynomial majorants* of indicator functions.

**Lemma 8.2** (Majorant certificate). *Let  $C, a, b, c, r \in \mathbb{Q}$  and suppose*

$$\varphi(y) := C|y|^3 + ay + by^2 + c$$

*satisfies  $\varphi \geq 0$  on  $\mathbb{R}$  and  $\varphi \geq 1$  on  $(-\infty, r]$ . Then for every admissible  $\mu$ ,*

$$\mathbb{P}(X \leq r) \leq \mathbb{E}\varphi(X) = C\beta + b + c,$$

*using  $\mathbb{E}X = 0$  and  $\mathbb{E}X^2 = 1$ .*

*Proof.*  $\mathbf{1}_{(-\infty, r]} \leq \varphi$  pointwise, and  $\varphi$  is  $\mu$ -integrable by admissibility; the displayed value of  $\mathbb{E}\varphi$  is immediate.  $\square$

*In Lean:* `prob_le_le_integral, integral_cubic_majorant, prob_le_le_of_certificate` in `Upper0423/SingleSummand.lean`.

The hypotheses of Lemma 8.2 are themselves certified by two sum-of-squares identities, which is what makes them checkable by `nlinarith` on rational data.

**Lemma 8.3** (SOS1). *Let  $C > 0$  and  $v \in \mathbb{R}$  with  $b + 2Cv > 0$  and  $(Cv^2 - a)^2 \leq 4(b + 2Cv)c$ . Then  $Cz^3 + bz^2 + az + c \geq 0$  for all  $z \geq 0$ .*

*Proof.* Multiply by  $4(b + 2Cv) > 0$  and use the identity

$$\begin{aligned} 4(b + 2Cv)(Cz^3 + bz^2 + az + c) &= 4C(b + 2Cv)z(z - v)^2 \\ &\quad + (2(b + 2Cv)z - (Cv^2 - a))^2 + (4(b + 2Cv)c - (Cv^2 - a)^2). \end{aligned}$$

Each of the three summands is nonnegative.  $\square$

**Lemma 8.4** (SOS2). *Let  $C > 0$  and  $b^2 \leq -3Ca$ . Then  $z \mapsto Cz^3 + bz^2 - az + c$  is nondecreasing.*

*Proof.* For  $w \leq z$ ,

$$\begin{aligned} 12C \Big[ (Cz^3 + bz^2 - az + c) - (Cw^3 + bw^2 - aw + c) \Big] \\ = (z - w) \Big( 3C^2(z - w)^2 + (3C(z + w) + 2b)^2 + 4(-3Ca - b^2) \Big) \geq 0. \end{aligned}$$

$\square$

*In Lean:* `cubic_nonneg, cubic_mono, certificate_nonpos, certificate_last` in `Upper0423/SingleSummand.lean`.

*Proof of Theorem 8.1.* By replacing  $X$  with  $-X$  it suffices to prove the one-sided bound  $\mathbb{P}(X \leq t) - \Phi(t) \leq \frac{21}{50}\beta$  for all  $t$  (the reverse discrepancy is obtained from the one-sided bound for  $-X$  together with  $\Phi(-t) = 1 - \Phi(t)$ , and  $\beta$  is unchanged). Recall  $\beta \geq 1$  (Lemma 4.1). The range of  $t$  is split into ten pieces:

- $t \leq -6/5$ : a pure quadratic majorant ( $C = 0$ ) with  $(a, b, c) = (-\frac{5}{3} \cdot \frac{900}{3721}, \frac{900}{3721}, \frac{25}{36} \cdot \frac{900}{3721})$  and  $r = -6/5$  gives  $\mathbb{P}(X \leq t) \leq b + c$ , which is below  $\frac{21}{50}\beta$  already since  $\Phi \geq 0$ .
- $-6/5 < t \leq -1$ : another pure quadratic majorant with  $(a, b, c) = (-\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ ,  $r = -1$ , combined with the explicit Gaussian lower bound of Lemma 8.5 at  $z = 6/5$ .

- seven cells with endpoints  $-1, -\frac{9}{10}, -\frac{4}{5}, -\frac{3}{5}, -\frac{2}{5}, -\frac{1}{5}, 0, \frac{3}{10}$ : each carries a cubic certificate with  $C = \frac{21}{50}$  and explicit rational  $(a, b, c, v)$ , verified by Lemmas 8.3 and 8.4, together with the Gaussian value at the left endpoint of the cell.
- $t > 3/10$ :  $\mathbb{P}(X \leq t) \leq 1$  and  $\Phi(t) \geq \Phi(3/10) \geq \frac{1}{2} + \frac{197}{1680}$ , which suffices.

In each of the seven cells the estimate has the shape

$$\mathbb{P}(X \leq t) - \Phi(t) \leq \underbrace{\mathbb{P}(X \leq r) - \Phi(l)}_{\leq C\beta + b + c} - \underbrace{\Phi(l)}_{\geq b + c} \leq C\beta,$$

where  $[l, r]$  is the cell, using monotonicity of both  $F$  and  $\Phi$ ; the numerical content of each row is the pair of inequalities  $b + c \leq \Phi(l)$  and the SOS conditions.  $\square$

*In Lean:* `prob_le_sub_normalCDF_le`, `abs_prob_le_sub_normalCDF_le`, `normalizedError_one_le` in `Upper0423/SingleSummand.lean`.

**Lemma 8.5** (Explicit Gaussian bounds). *For  $z \geq 0$ ,  $\Phi(-z) \geq \frac{1}{2} - \frac{2}{5}(z - \frac{z^3}{6} + \frac{z^5}{40})$ , and  $\Phi(3/10) \geq \frac{1}{2} + \frac{197}{1680}$ .*

*Proof.* From  $e^{-v} \leq 1 - v + \frac{v^2}{2}$  for  $v \geq 0$  one gets  $\phi(x) \leq \frac{2}{5} - \frac{x^2}{5} + \frac{x^4}{20}$ ; integrating over  $[-z, 0]$  gives the first bound. From  $1 - \frac{x^2}{2} \leq e^{-x^2/2}$  and the numerical bound  $(2\pi)^{-1/2} \geq 25/63$  one gets  $\phi(x) \geq \frac{25}{63}(1 - \frac{x^2}{2})$  on  $[0, 3/10]$ ; integrating gives the second.  $\square$

*In Lean:* `exp_neg_le_quadratic`, `gaussianPDFReal_le_quartic`, `normalCDF_neg_ge`, `normalCDF_three_tenths_ge` in `Upper0423/SingleSummand.lean`.

## 9 The large Lyapunov fraction

**Theorem 9.1.** *For every admissible  $\mu$ , every  $n \geq 1$  and every  $x$ ,  $|F_n(x) - \Phi(x)| \leq \frac{11}{20}$ . Consequently, if  $\ell = \beta/\sqrt{n} \geq 13003/10000$ , then*

$$\mathcal{E}(\mu, n, x) = \frac{\sqrt{n}}{\beta} |F_n(x) - \Phi(x)| \leq \frac{10000}{13003} \cdot \frac{11}{20} = \frac{5500}{13003} < \frac{423}{1000}.$$

*Proof.*  $S_n$  has mean 0 and variance 1, so Cantelli's (one-sided Chebyshev) inequality gives  $\mathbb{P}(S_n \geq x) \leq \frac{1}{1+x^2}$  for  $x > 0$  and hence  $F_n(x) \geq \frac{x^2}{1+x^2}$ ; symmetrically  $F_n(x) \leq \frac{1}{1+x^2}$  for  $x < 0$ . For  $x \geq 0$  one then needs the elementary Gaussian gap estimate

$$\Phi(x) - \frac{x^2}{1+x^2} \leq \frac{11}{20} \quad (x \geq 0),$$

proved in `GaussianBounds.lean` by a polynomial argument on  $[0, 1]$  and by monotonicity beyond, together with  $F_n \leq 1$  and  $\Phi \geq \frac{1}{2}$ ; the case  $x < 0$  is symmetric.  $\square$

*In Lean:* `cantelli_upper`, `cantelli_lower`, `normalCDF_cantelli_gap`, `cdf_error_le_eleven_twentieths`, `normalizedSum_cdf_error_le_eleven_twentieths` in `BerryEsseen/Cantelli.lean`, `BerryEsseen/GaussianBounds.lean`.

*In Lean:* `normalizedError_large_ell` in `Upper0423/LargeEll.lean`.

**Remark 9.2.** The threshold is determined by the numerical identity  $\frac{11}{20} \cdot \frac{10000}{13003} = 0.4229793 \dots < 0.423$ : any smaller threshold would fail, and  $13003/10000$  is the value the finite covers are built for. A sharper unconditional cap than  $11/20$  would lower the threshold and shrink the finite work; conversely, one cannot reuse a threshold calibrated for a weaker target, e.g.  $56/45 = 1.24$ , because  $\frac{11}{20} \cdot \frac{45}{56} = 0.4419 \dots > 0.423$ .

## 10 The interval and Taylor-jet engine

The remaining two regions require numerical evaluation of the four integrals of Definition 5.2. This section describes the machinery: an executable fixed-point interval arithmetic with fourth-order Taylor jets, a quadrature rule, a majorant assembly, and the Lean theorems that make the whole thing sound. Nothing in this section is trusted; everything is proved.

### 10.1 Fixed-point intervals

**Definition 10.1.** Let  $S = 2^{40} = 1099511627776$ . An *interval* is a pair of integers  $[\ell, h]$ , and it *contains*  $x \in \mathbb{R}$  when  $\ell \leq Sx \leq h$ .

*In Lean:* `Iv`, `Iv.Contains`, `scale`, `precision` in `Upper0423/Native/Interval.lean`, `Upper0423/Native/IntervalSound.lean`.

All arithmetic is exact integer arithmetic with outward rounding, so containment is preserved. Writing  $\lfloor \cdot \rfloor$  and  $\lceil \cdot \rceil$  for floor/ceiling division of integers (defined for divisors of either sign), the operations are

|                                |                                                                                                               |
|--------------------------------|---------------------------------------------------------------------------------------------------------------|
| negation                       | $[-h, -\ell];$                                                                                                |
| addition                       | $[\ell_1 + \ell_2, h_1 + h_2];$                                                                               |
| subtraction                    | $[\ell_1 - h_2, h_1 - \ell_2];$                                                                               |
| multiplication                 | $[\lfloor \min(\dots)/S \rfloor, \lceil \max(\dots)/S \rceil]$ over the four endpoint products;               |
| division by $n \in \mathbb{N}$ | $[\lfloor \ell/n \rfloor, \lceil h/n \rceil];$                                                                |
| reciprocal                     | $[\lfloor S^2/h \rfloor, \lceil S^2/\ell \rceil]$ (fails if $\ell \leq 0 \leq h$ );                           |
| square                         | as multiplication, with lower bound 0 if $\ell \leq 0 \leq h$ ;                                               |
| square root                    | $[\lfloor \sqrt{\ell S} \rfloor, \lceil \sqrt{h S} \rceil]$ via <code>Nat.sqrt</code> (fails if $\ell < 0$ ). |

Each of these has a soundness lemma of the form “if  $a$  contains  $x$  and  $b$  contains  $y$  then  $a * b$  contains  $x * y$ ”. Exponentiation to a natural power is by the same binary-exponentiation loop as the reference C++ implementation, so that the Lean checker and the search program agree bit for bit.

### 10.2 Transcendental seeds

**Lemma 10.2** ( $\pi$ ). *The interval  $[3454217652352, 3454217652372]$  contains  $\pi$ .*

*Proof.* Immediate from `Mathlib’s Real.pi_gt_d20` and `Real.pi_lt_d20`. □

*In Lean:* `piIv`, `contains_piIv` in `Upper0423/Native/TranscendentalSound.lean`.

**Lemma 10.3** (exponential). *The routine `Iv.exp` returns an interval containing  $e^x$  whenever it succeeds.*

*Proof sketch.* Endpointwise. For  $x \leq -40S$  return  $[0, 1]$ , which is valid because  $e^{-40} \leq 1/S$ . Otherwise find the least  $k$  with  $8|x| \leq S2^k$ , put  $y = x/2^k$  (so  $|y| \leq S/8$ ), sum twelve Taylor terms of  $e^y$  by the recursion  $(\Sigma, T) \mapsto (\Sigma + Ty/(j+1), Ty/(j+1))$ , widen the result by one unit in each direction to absorb the truncation error, clamp at 0, and square  $k$  times. The Taylor remainder after twelve terms at  $|y| \leq 1/8$  is far below one unit in the last place. □

**Lemma 10.4** (sine and cosine). *The routine `Iv.sinCos` returns a pair of intervals containing  $(\sin x, \cos x)$  for every  $x$  in the argument interval.*

*Proof sketch.* A point evaluation shifts the argument by the nearest multiple of  $\pi/2$  into  $[-1, 1]$  (using the enclosure of Lemma 10.2), evaluates degree-17 Taylor polynomials, widens by one unit, and permutes/negates according to the quadrant. For an interval argument, the two endpoint enclosures are hulled, and additionally hulled with  $\pm 1$  at every stationary point ( $k\pi$  for  $\cos$ ,  $(2k+1)\pi/2$  for  $\sin$ ,  $|k| \leq 4$ ) whose own enclosure meets the argument interval; the result is intersected with  $[-1, 1]$ . Soundness rests on the observation that between two consecutive stationary points the function is monotone, so its range over a subinterval is spanned by the endpoint values together with the extreme values at any interior stationary point. Arguments of absolute value larger than 7 return  $[-1, 1]$ .  $\square$

*In Lean:* `expEndpoint_sound`, `Contains.exp`, `sinCosEndpoint_sound`, `Contains.sinCos` in `Upper0423/Native/TranscendentalSound.lean`, `Upper0423/Native/TrigSound.lean`.

### 10.3 Fourth-order jets

**Definition 10.5.** A *jet* is a quintuple of intervals  $(c_0, \dots, c_4)$ . It *encloses* a function  $F$  at a point  $\sigma$  when  $c_k$  contains  $F^{(k)}(\sigma)/k!$  for  $k = 0, \dots, 4$ . It encloses  $F$  on a set  $D$  when  $F$  is four times differentiable on  $D$  and the enclosure holds at every point of  $D$ .

*In Lean:* `Jet`, `Jet.Encloses`, `Jet.EnclosesOn` in `Upper0423/Native/JetSound.lean`.

Jets are propagated by the usual normalized-coefficient recurrences: products by the Cauchy convolution, reciprocal and square root by the Newton-type recurrences, exponential and sine/cosine by the weighted convolutions  $r_k = \frac{1}{k} \sum_{j=1}^k j a_j r_{k-j}$  (and its trigonometric pair). The zeroth coefficient of a squaring uses the dedicated interval square (so that  $c_0 \geq 0$  is exploited). Each of these has a Lean soundness lemma, reducing to scale-free coefficient identities in the `BerryEsseen.Numerics` namespace; only interval containment is redone in the engine's own namespace.

### 10.4 The quadrature rule

A *panel* is a rational interval  $[l, r]$ . Parametrize it by  $t = m + h\sigma$  with  $m = \frac{l+r}{2}$ ,  $h = \frac{r-l}{2}$  and  $\sigma \in [-1, 1]$ .

**Lemma 10.6** (Panel rule). *Let  $F$  be a real function,  $l < r$  rationals. Suppose a jet  $W$  encloses  $\sigma \mapsto F(m + h\sigma)$  on  $[-1, 1]$  and a jet  $M$  encloses it at  $\sigma = 0$ . Then*

$$\int_l^r F(t) dt \leq \min \left\{ (r-l) \left( \overline{M_0} + \frac{\overline{M_2}}{3} + \frac{\overline{W_4}}{5} \right), (r-l) \overline{W_0} \right\},$$

where  $\overline{\cdot}$  denotes the upper endpoint of an interval, rescaled by  $S^{-1}$ .

*Proof sketch.* For the second (range) bound,  $F \leq \overline{W_0}$  on the panel. For the first (Taylor) bound, expand  $F(m + h\sigma) = \sum_{k \leq 3} c_k \sigma^k + c_4(\xi) \sigma^4$  with  $c_k = c_k(0)$  for  $k \leq 3$  the midpoint coefficients and  $c_4(\xi)$  a fourth coefficient at some interior point. Integrating  $\sigma^k$  over  $[-1, 1]$  kills the odd powers and gives  $\int_{-1}^1 \sigma^2 = \frac{2}{3}$ ,  $\int_{-1}^1 \sigma^4 = \frac{2}{5}$ ; multiplying by  $h = \frac{r-l}{2}$  yields the stated combination  $(r-l)(c_0 + \frac{c_2}{3} + \frac{c_4}{5})$ , and the whole-panel jet supplies the uniform upper bound for  $c_4$ .  $\square$

*In Lean:* `panelUpper`, `integral_le_panelUpper`, `affineJet`, `centerJet` in `Upper0423/Native/Majorant.lean`, `Upper0423/Native/PanelSound.lean`.

The rule is a genuine upper bound for the integral, not an approximation. The fourth-order Taylor form is what makes the certificates small: on a smooth panel it converges like  $(r-l)^5$  while the range rule converges only like  $(r-l)$ .

## 10.5 Panel majorants

A certificate leaf fixes a *box*

$$b = (n, \beta_{\text{lo}}, \beta_{\text{hi}}, \eta_{\text{lo}}, \eta_{\text{hi}}, c, \tau, \varepsilon)$$

of rational data, subject to the validity conditions

$$n \geq 2, \quad 1 \leq \beta_{\text{lo}} \leq \beta_{\text{hi}}, \quad 0 \leq \eta_{\text{lo}} \leq \eta_{\text{hi}} \leq 1, \quad c > 0, \quad 0 < \varepsilon < \tau \leq \frac{1}{2}, \quad c\tau < \frac{3}{2}, \quad c < \pi.$$

The derived quantities used by the majorants are

$$\begin{aligned} \delta &= \beta_{\text{hi}} - 1, & d &= \min\left(1 - \frac{1}{\beta_{\text{hi}}}, 1 - \eta_{\text{lo}}\right), & a &= \beta_{\text{hi}} + \eta_{\text{hi}}, \\ \Sigma_1 &= \delta(\beta_{\text{hi}} + \frac{5}{3}), & \Sigma_2 &= \beta_{\text{hi}}^2 - \eta_{\text{lo}}^2, & J &= \frac{1}{36} \min(\Sigma_1, \Sigma_2). \end{aligned}$$

The reason  $d$  is a legitimate upper bound for  $1 - s$  is the pair of inequalities  $s \geq 1/\beta$  (from  $s\beta \geq 1$ , Lemma 4.2) and  $s \geq \eta$ ; the quantities  $\Sigma_1, \Sigma_2$  dominate  $(\beta - 1)(\beta + 5/3)$  and  $\beta^2 - \eta^2$  respectively.

*In Lean:* `Box`, `Box.valid`, `Box.dcap`, `Box.dsum`, `Box.cubic1`, `Box.cubic2`, `Box.imagSq`, `dcap_ge`, `dsum_ge`, `cubic1_ge`, `cubic2_ge` in `Upper0423/Native/Majorant.lean`, `Upper0423/Analytic/ModulusBridge.lean`.

With  $u = ct$ , the four *branch choices* of a panel are as follows.

**Imaginary majorant**  $j(u)$ . One of

$$\frac{1}{6}\sqrt{\Sigma_1}u^3, \quad \frac{1}{2}\sqrt{\frac{2}{3}}\delta u^2, \quad \frac{1}{6}\sqrt{\Sigma_2}u^3, \quad \sqrt{2d}(\sin u - u \cos u) + d u^2,$$

dominating  $|\text{Im } f(u)|$  by Propositions 6.8, 6.9, 6.10, 6.11 respectively.

**Real majorant**  $\text{re}(u)$ . One of

$$\cos u + \frac{1}{6}\delta u^3, \quad \cos u + d(u \sin u + u^2), \quad 1 - \frac{u^2}{2} + \kappa\beta_{\text{hi}}u^3,$$

dominating  $\text{Re } f(u)$  by Proposition 6.7.

**Modulus majorant**  $M(u)$ . One of

$$\begin{aligned} \text{one:} & \quad 1; \\ \text{line:} & \quad \sqrt{1 - u^2 + 2\kappa a u^3}; \\ \text{cosine:} & \quad \sqrt{1 - \frac{2A(1 - \cos(au))}{a^2}}; \\ \text{coordinate:} & \quad \sqrt{\text{far}(\cos u, \text{re}(u); G) + j(u)^2}. \end{aligned}$$

The first three are  $|f| \leq 1$  and Propositions 6.2, 6.5; the *cosine* branch is guarded by a whole-panel check that  $4 \leq a c l$  and  $a c r \leq 2\pi$ , so that the curved branch of the profile applies on the entire panel. The *coordinate* branch is the disk/strip intersection:  $|f|^2 = (\text{Re } f)^2 + (\text{Im } f)^2$ , the imaginary square is bounded by  $j(u)^2$ , and the real square is bounded by

$$\text{far}(\alpha, \gamma; G) = \begin{cases} \gamma^2, & G = +1, \\ \alpha^2, & G = -1, \\ \alpha^2 + \gamma^2, & G = 0, \end{cases}$$

where  $\alpha \leq \operatorname{Re} f(u) \leq \gamma$  are the lower bound  $\cos u$  (Proposition 6.6) and the chosen upper bound, and  $G$  is a *sign guard* computed once on the whole panel:  $G = +1$  if  $\alpha + \gamma \geq 0$  throughout the panel (so  $|\operatorname{Re} f| \leq \gamma$ ),  $G = -1$  if  $\alpha + \gamma \leq 0$  throughout (so  $|\operatorname{Re} f| \leq -\alpha$ ), and  $G = 0$  otherwise. Since  $\max(|\alpha|, |\gamma|)^2 \leq \alpha^2 + \gamma^2$  always, the guard only improves the bound, and its correctness on the whole panel is what allows the sharper midpoint jet to reuse it. The idea of intersecting the Taylor disk with the real and imaginary strips, which is what this branch implements, is due to [XL] (item (vi) of Section 2); the sign guard and the use of the two-sided real bracket of Proposition 6.6 are particular to the present engine.

*In Lean:* `imagJet`, `realJet`, `modulusJet`, `cosineGuard`, `signGuard`, `farthestSquare`, `abs_im_le_imagR`, `re_le_realR`, `norm_le_modulusR`, `cosineGuard_sound` in `Upper0423/Native/Majorant.lean`, `Upper0423/Analytic/ModulusBridge.lean`, `Upper0423/Analytic/IntegrandBridge.lean`.

## 10.6 The low-frequency error factor

The low-frequency integrand (13) needs a bound for  $|f(u)^n - g(u)^n|$ , and this is where the geometric-sum comparison enters. Writing  $G_n(y, z) = \sum_{j < n} y^j z^{n-1-j}$ , the telescoping identity gives

$$|f(u)^n - g(u)^n| \leq |f(u) - g(u)| \cdot G_n(M(u), g(u)) \quad (19)$$

whenever  $|f| \leq M$  and  $g \geq 0$ . The engine computes  $G_n$  exactly by the halving recursion

$$G_{2k}(y, z) = (y^k + z^k) G_k(y, z), \quad G_{2k+1}(y, z) = y G_{2k}(y, z) + z^{2k},$$

returning  $(y^n, z^n, G_n)$  from  $(y^{\lfloor n/2 \rfloor}, z^{\lfloor n/2 \rfloor}, G_{\lfloor n/2 \rfloor})$ ; this costs  $O(\log n)$  jet multiplications.

*In Lean:* `geometric`, `geometricSum`, `geomSum`, `geomSum_double`, `geomSum_succ_double`, `geometricR_spec`, `norm_pow_sub_pow_le_geomSum` in `Upper0423/Native/Majorant.lean`, `Upper0423/Analytic/Geometric.lean`.

Five branches for the factor  $\Lambda(u)$  with  $|f(u)^n - g(u)^n| \leq u^3 \Lambda(u)$  (or the trivial split) are provided:

- *classical*:  $\Lambda = (\frac{1}{6}\beta_{\text{hi}} + \text{GU}(u)) G_n(M, g)$ , where  $g(u) - 1 + \frac{u^2}{2} \leq u^3 \text{GU}(u)$  and  $\|f(u) - 1 + \frac{u^2}{2}\| \leq \frac{1}{6}\beta|u|^3$  by (18): the two Taylor disks around the common center  $1 - \frac{u^2}{2}$ .
- *coordinate*: the real and imaginary parts of  $f(u) - g(u)$  are bracketed separately. Using the two-sided Taylor brackets  $u^3 \text{GC}_L(u) \leq g(u) - \cos u \leq u^3 \text{GC}_U(u)$  one gets  $\alpha' \leq \operatorname{Re}(f(u) - g(u)) \leq \gamma'$  with  $\alpha' = -\text{GC}_U(u)u^3$  and  $\gamma' = (\frac{\delta}{6} - \text{GC}_L(u))u^3$ , so that  $\Lambda = \sqrt{\text{far}(\alpha', \gamma'; G')} + J \cdot G_n(M, g)$  with a second sign guard  $G'$  and the imaginary square cap  $J$ .
- *anchor*: split around  $\cos u$ :  $\Lambda = \sqrt{\frac{\delta^2}{36}} + J G_n(M, \cos u) + \text{GC}_U(u) G_n(g, \cos u)$ .
- *ode*: the same split with the ODE bound of Proposition 6.12 in place of the anchor disk:  $\Lambda = \sqrt{\Sigma_1} \text{SU}(u) G_n(M, \cos u) + \text{GC}_U(u) G_n(g, \cos u)$ , where  $u - \sin u \leq u^3 \text{SU}(u)$ .
- *trivial*:  $|f^n - g^n| \leq M^n + g^n$ , used only where the difference bound is worthless.

The polynomials  $\text{GU}, \text{GC}_L, \text{GC}_U, \text{SU}$  are the explicit alternating Taylor truncations evaluated by the engine, and the corresponding real inequalities are proved on  $0 \leq u \leq \pi/2$  in `Analytic/TaylorPolys.lean` from Mathlib's exponential and trigonometric Taylor bounds.

*In Lean:* `lowFactor`, `lowerErrorJet`, `upperErrorJet`, `gaussianCosineCubic`, `sineCubicUpper`, `gaussianCubicUpper` in `Upper0423/Native/Majorant.lean`.

*In Lean:* `gaussianCubicUpper_bound`, `gaussianCosine_upper_bound`, `gaussianCosine_lower_bound`, `sineCubicUpper_bound`, `norm_pow_sub_gauss_le_lowFactor`, `low_integrand_le`, `high_integrand_le`, `normal_integrand_le` in `Upper0423/Analytic/TaylorPolys.lean`, `Upper0423/Analytic/IntegrandBridge.lean`.

## 10.7 The panel integrand and its certified bound

Putting the pieces together, the engine's real majorant for a panel  $[l, r]$  of part  $p \in \{\text{low, high, normal}\}$  with branch choices and guards is

$$\text{build}_b(t) = \begin{cases} |K(t)|_{\text{normal}} \cdot e^{-nc^2t^2/2}, & p = \text{normal}, \\ |K(t)|_{\text{left/right}} \cdot M(ct)^n, & p = \text{high}, \\ c^3t^2 \cdot (s|K(s)|)|_{s=t} \cdot \Lambda(ct), & p = \text{low}, \\ |K(t)|_{\text{left}} \cdot (M(ct)^n + e^{-c^2t^2/2 \cdot n}), & p = \text{low, trivial}. \end{cases}$$

(The low case uses form (9):  $c^3t^2 \cdot s|K(s)| = u^3|K(t)|$  with  $u = ct$ .) Comparing with (13), each case dominates the corresponding Prawitz integrand pointwise on the panel, by the results of Sections 6, 10.5 and 10.6. The function `framePanel` then

- (1) checks the box validity, the panel validity ( $\varepsilon \leq l < r \leq \tau$  for low,  $\tau \leq l < r \leq 1$  and no straddling of  $\frac{1}{2}$  for high,  $0 \leq l < r \leq \tau$  for normal) and the branch validity;
- (2) evaluates the jet of `buildb` on the whole panel, *computing* the sign guards from it;
- (3) evaluates the jet at the midpoint, *reusing* the stored guards;
- (4) returns the panel rule value of Lemma 10.6, or fails.

**Theorem 10.7** (Panel soundness). *If `framePanel bl r p ch` = some  $U$ , then  $b$  and the panel are valid,  $U \geq 0$ , and there are guards  $g$  valid on all of  $[l, r]$  with*

$$\int_l^r \text{build}_b(t) dt \leq U.$$

*If moreover  $\mu$  is admissible with  $(\beta, \eta)$  in the box, then*

$$\int_l^r (\text{the corresponding Prawitz integrand}) dt \leq U.$$

*In Lean:* `framePanel_sound`, `framePanel_sound'`, `lowPanel_integral_le`, `highPanel_integral_le`, `normalPanel_integral_le` in `Upper0423/Native/PanelSound.lean`, `Upper0423/Small/Chains.lean`, `Upper0423/Small/Leaf.lean`.

## 10.8 Chains, omission and tail

A *chain* is a list of adjacent panels covering an interval; the chain total bounds the integral over the whole interval, provided the endpoints match, which the checker verifies explicitly (the low chain must end exactly at  $\tau$ , the high chain at 1, the normal chain at  $\tau$ ).

*In Lean:* `chainSum`, `chainSum_integral_le` in `Upper0423/Native/Cover.lean`, `Upper0423/Small/Chains.lean`.

The interval  $[0, \varepsilon]$  of the low part is not covered by panels: the kernel is singular at 0 and the Taylor jets degenerate. Instead it is bounded in closed form.

**Lemma 10.8** (Low-frequency omission). *Let  $0 < \varepsilon \leq \frac{1}{2}$ ,  $c > 0$ ,  $n \geq 1$  and  $\beta \leq \beta_{\text{hi}}$ . Then*

$$\int_0^\varepsilon |K(t)| |f(ct)^n - g(ct)^n| dt \leq n \left[ \beta_{\text{hi}} c^3 \left( \frac{\varepsilon^4}{24} + \frac{\varepsilon^3}{18\pi} \right) + c^4 \left( \frac{\varepsilon^5}{40} + \frac{\varepsilon^4}{32\pi} \right) \right].$$

*Proof sketch.* For  $0 < t \leq \frac{1}{2}$  one has  $\cot(\pi t) \leq \frac{1}{\pi t}$ , whence  $0 \leq k(t) \leq \frac{1}{\pi t}$  and  $|K(t)| \leq 1 + \frac{1}{\pi t}$ . Also  $|f^n - g^n| \leq n|f - g| \leq n(\frac{\beta u^3}{6} + \frac{u^4}{8})$  with  $u = ct$ , by the two Taylor disks around  $1 - \frac{u^2}{2}$ . Multiplying and integrating  $t^2, t^3, t^4$  over  $[0, \varepsilon]$  gives the stated expression.  $\square$

*In Lean:* norm\_prawitzKernel\_le\_one\_add, omission\_bound, omissionUpper, omissionUpper\_sound in Upper0423/Small/Omission.lean, Upper0423/Native/Cover.lean.

The Gaussian tail (14) is handled by panelling  $v \mapsto e^{-v}/v$  on  $[x, x + 24]$  with deterministic cut points (doubling from  $x$  until  $\frac{1}{4}$ , then steps of  $\frac{1}{4}$ ), plus the closed tail bound  $\int_R^\infty e^{-v}/v dv \leq e^{-R}/R$ .

*In Lean:* tailCuts, tailChain, tailUpper, tailPanel\_sound, tailChain\_sound, integral\_exp\_neg\_div\_Ioi\_le, tailUpper\_sound in Upper0423/Native/Cover.lean, Upper0423/Small/Tail.lean, Upper0423/Native/PanelSound.lean.

## 10.9 A leaf and its budget

**Definition 10.9.** Given a box  $b$  and leaf data  $d$  (the cutoff  $c$ , the split  $\tau$  and the three chains), the *leaf total* is

$$U(b, d) = \underbrace{\Sigma_{\text{low}}}_{[\varepsilon, \tau]} + \underbrace{\Sigma_{\text{high}}}_{[\tau, 1]} + \underbrace{\Sigma_{\text{normal}}}_{[0, \tau]} + \underbrace{\Omega}_{[0, \varepsilon]} + \underbrace{\mathcal{T}}_{\text{tail}},$$

and the leaf is *accepted* when  $b$  is valid,  $U \geq 0$  and

$$nU^2 < \left(\frac{423}{1000}\right)^2 \beta_{\text{lo}}^2. \quad (20)$$

*In Lean:* leafTotal, leafAccepted in Upper0423/Native/Cover.lean.

**Theorem 10.10** (Leaf soundness). *Let  $\mu$  be admissible with  $\beta_{\text{lo}} \leq \beta \leq \beta_{\text{hi}}$  and  $\eta_{\text{lo}} \leq \eta \leq \eta_{\text{hi}}$ , and suppose the leaf  $(b, d)$  is accepted. Then for every  $x$ ,*

$$\mathcal{E}(\mu, n, x) \leq \frac{423}{1000}.$$

*Proof.* Theorem 5.3 with  $T = c\sqrt{n}$  and split  $\tau$  bounds  $|F_n(x) - \Phi(x)|$  by the smoothing integral; splitting the low integral at  $\varepsilon$  and applying the chain, omission and tail bounds gives  $|F_n(x) - \Phi(x)| \leq U$ . The budget (20) is equivalent to  $\sqrt{n}U < \frac{423}{1000}\beta_{\text{lo}}$  (both sides nonnegative), hence

$$\mathcal{E}(\mu, n, x) = \frac{\sqrt{n}}{\beta} |F_n - \Phi| \leq \frac{\sqrt{n}U}{\beta} \leq \frac{423}{1000} \cdot \frac{\beta_{\text{lo}}}{\beta} \leq \frac{423}{1000},$$

using  $\beta_{\text{lo}} \leq \beta$ . □

*In Lean:* leafTotal\_cdf\_bound, leafAccepted\_normalizedError in Upper0423/Small/Leaf.lean.

Note the shape of (20): it is a purely rational inequality in the leaf data, which is what makes it decidable, and the square is taken to avoid  $\sqrt{n}$  inside the checker.

## 11 The per-sample-size covers, $2 \leq n \leq 255$

### 11.1 Cover trees

**Definition 11.1.** A *cover* is a binary tree whose leaves carry leaf data and whose internal nodes carry an axis and a rational split point. Acceptance over a root rectangle is defined recursively: a leaf is accepted if the leaf box obtained by outward rounding of the rectangle to the  $2^{-40}$  grid is accepted in the sense of Definition 10.9; a split node is accepted if the split point lies strictly inside the rectangle along the chosen axis and both children are accepted over the two halves.

*In Lean:* Cover, coverAccepted, coverAcceptedBE, roundDown, roundUp, imageBox, imageBoxBE in Upper0423/Native/Cover.lean, Upper0423/Native/CoverGen.lean.

**Lemma 11.2** (Locating a law in a cover). *If a cover is accepted over a rectangle  $R$  and  $(\beta, \eta)$  (respectively  $(\delta, w)$ ) lies in  $R$ , then some accepted leaf box contains  $(\beta, \eta)$ .*

*Proof.* Induction on the tree: at a split node, one of the two halves contains the point because the split point lies inside the rectangle; at a leaf, outward rounding only enlarges the box.  $\square$

*In Lean:* `coverAccepted_leaf`, `cov_coverAcceptedBE_leaf`, `inBox_imageBox`, `cov_inBox_imageBoxBE` in `Upper0423/Small/CoverTree.lean`, `Upper0423/Covers/Sound.lean`.

Two coordinate systems are used, as anticipated in Section 4. For  $\beta \leq 2$ , the cover is a tree in  $(\delta, w) \in [\delta_{\text{lo}}, \delta_{\text{hi}}] \times [0, 1]$  with  $\beta = 1 + \delta$  and  $\eta = 1 - \delta w$ ; the map  $(\delta, w) \mapsto (\beta, \eta)$  is well defined because, given  $\delta \geq 0$  and  $\eta \in [\max(0, 1 - \delta), 1]$ , the value  $w = (1 - \eta)/\delta$  (or  $w = 0$  if  $\delta = 0$ ) lies in  $[0, 1]$  and satisfies  $1 - \delta w = \eta$ ; here  $\eta \geq 1 - \delta$  is exactly Proposition 4.3. For  $\beta \geq 2$  the cover is a tree directly in  $(\beta, \eta) \in [\beta_{\text{lo}}, \beta_{\text{hi}}] \times [0, 1]$ .

## 11.2 The certificate format

Each cover is serialized as a whitespace-separated token string and parsed by an untrusted recursive-descent parser: `S axis m <left> <right>` for a split and

`F c tau nL (r code)* nH (r code)* nN r*`

for a leaf, where `code` packs the four branch choices into a single natural number (imag + 4 real + 12 modulus + 48 error). Parsing is untrusted precisely because the soundness theorem is universally quantified over *every* tree the checker accepts: if the parser produced garbage, the checker would simply reject it. A representative leaf, from `Covers/Data/BE3.lean`, reads

`F 477/200 2/5 17 1/500 216 1/250 216 ... 20 17/40 20 ... 13 2/65 4/65 ... 2/5`

declaring  $c = 477/200$ ,  $\tau = 2/5$ , 17 low panels, then the high panels and the 13 normal panel endpoints.

*In Lean:* `parseCover`, `tokens`, `coverStringAccepted`, `coverStringAcceptedDW`, `coverStringAcceptedBE` in `Upper0423/Native/Cover.lean`, `Upper0423/Native/CoverGen.lean`.

## 11.3 The certificates

Table 4 summarizes the finite data. The counts were obtained by direct inspection of the repository.

| Family                               | Files | Leaves | Root domain                                              |
|--------------------------------------|-------|--------|----------------------------------------------------------|
| <code>Small/Data/Cover{2..63}</code> | 62    | 5 053  | $\delta \in [0, \frac{258569}{454000}]$ , $w \in [0, 1]$ |
| <code>Covers/Data/DW{2..255}</code>  | 254   | 8 953  | $\delta \in [\delta_n, d_n]$ , $w \in [0, 1]$            |
| <code>Covers/Data/BE{3..255}</code>  | 253   | 4 839  | $\beta \in [2, \beta_n]$ , $\eta \in [0, 1]$             |
| total                                | 569   | 18 845 |                                                          |

Table 4: The per-sample-size certificates. Here  $\beta_n$  is a rational number with  $\beta_n \geq \frac{13003}{10000}\sqrt{n}$ ;  $\delta_n = \frac{258569}{454000}$  for  $n \leq 63$  and  $\delta_n = 0$  for  $n \geq 64$ ; and  $d_n = \min(1, \beta_n - 1)$  (so  $d_n = 1$  except  $d_2 = \frac{17}{20}$ ). Each file carries exactly one `native_decide`.

For each  $n$  with  $2 \leq n \leq 255$  the file `Covers/All.lean` contains a theorem `cover_n` of the form

$$\beta \leq \beta_n \implies \mathcal{E}(\mu, n, x) \leq \frac{423}{1000},$$

proved by the case distinction

- $\beta \leq \frac{712569}{454000}$  and  $n \leq 63$ : the original research cover `Small/Data/Cover` $n$ ;
- $\frac{712569}{454000} < \beta \leq 2$  (or  $\beta \leq 2$  when  $n \geq 64$ ): the cover `Covers/Data/DW` $n$ ;
- $\beta > 2$ : the cover `Covers/Data/BE` $n$  (absent for  $n = 2$ , where  $\beta \leq \beta_2 = 37/20 < 2$  makes the branch vacuous).

The hypothesis  $\beta \leq \beta_n$  is supplied from  $\ell = \beta/\sqrt{n} < \frac{13003}{10000}$  by the elementary `beta_le_of_ell`: if  $b \geq 0$  and  $(\frac{13003}{10000})^2 n \leq b^2$  then  $\sqrt{n} \cdot \frac{13003}{10000} \leq b$ , so  $\beta < b$ . The rational  $\beta_n$  is the smallest convenient rational above  $\frac{13003}{10000}\sqrt{n}$ ; e.g.  $\beta_2 = 37/20$ ,  $\beta_3 = 227/100$ ,  $\beta_{64} = 521/50$ ,  $\beta_{255} = 1039/50$ . Finally `normalizedError_moderate_le` does `interval_cases n` and applies `cover_n`.

*In Lean:* `beta_le_of_ell`, `cover_2 ... cover_255`, `normalizedError_moderate_le` in `Upper0423/Covers/All.lean`.

*In Lean:* `normalizedError_le_of_coverAccepted`, `normalizedError_small_le` in `Upper0423/Small/Leaf.lean`, `Upper0423/Small/All.lean`.

*In Lean:* `normalizedError_le_of_dwCover`, `normalizedError_le_of_beCover` in `Upper0423/Covers/Sound.lean`.

**Remark 11.3.** The three families overlap deliberately: the `Small` covers were produced first, for the narrow range  $\beta \leq B = 712569/454000$ , and were retained rather than regenerated. The reason for the separate treatment of  $\beta \leq 2$  and  $\beta \geq 2$  is that in the first regime  $\eta$  is pinned near 1 by  $\beta + \eta \geq 2$ , so the  $(\delta, w)$  coordinates are much better adapted; for  $\beta \geq 2$  the constraint is inactive and  $(\beta, \eta)$  is the natural chart.

## 12 The uniform certificate, $n \geq 256$

For large  $n$ , a per- $n$  cover is impossible. Instead the whole smoothing functional is reduced to five one-dimensional integrals in two scalar parameters, and a single certificate covers all  $n \geq 256$  simultaneously.

### 12.1 The two scalars

Fix the smoothing parameters

$$a = \beta + \eta, \quad L = \frac{a}{\sqrt{n}}, \quad h = \frac{1}{\sqrt{n}}, \quad T = \frac{2\pi}{L} = \frac{2\pi\sqrt{n}}{a}, \quad \tau = \frac{1}{4}. \quad (21)$$

With this choice, (11) holds with  $c = 2\pi/a$ , so the per-summand frequency (12) is  $u = 2\pi t/a$ , and

$$a u = 2\pi t, \quad n u^2 = \frac{(2\pi t)^2}{L^2}, \quad (n-1) u^2 = \frac{(1-h^2)(2\pi t)^2}{L^2}.$$

Thus every occurrence of  $n$ ,  $\beta$  and  $\eta$  in the integrands collapses into  $L$  and  $h$ .

**Definition 12.1.** With  $q$  the convex profile of Definition 6.3 and  $\psi(t) = (2\pi t)^2 q(2\pi t)$ , set

$$\begin{aligned} \text{high}(L) &= \int_{1/4}^1 |K(t)| e^{-\psi(t)/L^2} dt, \\ \text{norm}(L) &= \int_0^{1/4} \left| K(t) - \frac{i}{\pi t} \right| e^{-2\pi^2 t^2/L^2} dt, \quad \text{tail}(L) = \int_{1/4}^\infty \frac{e^{-2\pi^2 t^2/L^2}}{\pi t} dt, \\ E_1(L, h, t) &= \exp\left(-\frac{(1-h^2)(2\pi t)^2}{2L^2}\right), \quad E_2(L, h, t) = \exp\left(-\frac{(1-h^2)(2\pi t)^2(\frac{1}{2} - 2\pi\kappa t)}{L^2}\right), \\ \text{lowG}(L, h) &= \int_0^{1/4} |K(t)| \frac{(2\pi t)^3}{12L^3} (E_1 + E_2) dt, \quad \text{lowK}(L, h) = \int_0^{1/4} |K(t)| \frac{(2\pi t)^4}{12L^4} E_1 dt. \end{aligned}$$

In Lean: psiR, highF, normalF, tailF, lowE1, lowE2, lowG, lowK in Upper0423/Uniform/Defs.lean.

Note  $E_1 = g(u)^{n-1}$  and  $E_2 = m(u)^{n-1}$  with  $m(u) = e^{-u^2/2+\kappa au^3}$ , so these are exactly the two exponential envelopes the geometric sum produces.

**Proposition 12.2** (Uniform reduction). *For every admissible  $\mu$ , every  $n \geq 1$  and every  $x$ ,*

$$\mathcal{E}(\mu, n, x) \leq \frac{a}{\beta} \cdot \frac{\text{high}(L) + \text{norm}(L) + \text{tail}(L)}{L} + \frac{\Gamma}{\beta} \text{lowG}(L, h) + \frac{h}{\beta} \text{lowK}(L, h), \quad (22)$$

with  $a, L, h$  as in (21) and  $\Gamma$  as in (5).

*Proof sketch.* Apply Theorem 5.3 with  $T = 2\pi/L$  and  $\tau = \frac{1}{4}$  and bound the four terms.

*High.* By Proposition 6.5,  $|f(u)| \leq e^{-u^2 q(a|u|)}$ , hence  $|f(u)|^n \leq e^{-nu^2 q(2\pi t)} = e^{-\psi(t)/L^2}$ ; the deduction  $|f| \leq e^{-u^2 q}$  from  $|f|^2 \leq 1 - 2u^2 q$  uses  $1 - 2y \leq e^{-2y}$ .

*Low.* By the geometric-sum comparison and the pointwise estimates of Propositions 6.12, 6.2 one has, for  $0 \leq u \leq \pi/2$ ,

$$|f(u)^n - g(u)^n| \leq \frac{n\Gamma u^3}{12} (g(u)^{n-1} + m(u)^{n-1}) + \frac{nu^4}{12} g(u)^{n-1},$$

where  $\Gamma = \sqrt{(\beta - 1)^2 + \beta^2 - \eta^2}$  arises as the radius of the disk containing  $f(u) - \cos u$  jointly with the imaginary strip. The constraint  $u \leq \pi/2$  holds on  $t \leq \frac{1}{4}$  because  $u = 2\pi t/a \leq \pi/(2a) \leq \pi/4$  by Proposition 4.3. Multiplying by  $|K(t)|$ , integrating, and substituting  $u = 2\pi t/a$  gives  $\frac{\Gamma}{\sqrt{n}} \text{lowG} + \frac{1}{n} \text{lowK}$ .

*Normal and tail.* These do not involve  $\mu$  at all and are literally  $\text{norm}(L)$  and  $\text{tail}(L)$ .

Finally  $\frac{\sqrt{n}}{\beta} = \frac{a}{\beta} \cdot \frac{1}{L}$  and  $\frac{\sqrt{n}}{\beta} \cdot \frac{1}{n} = \frac{h}{\beta}$ , which produces (22).  $\square$

In Lean: red\_high\_pointwise, red\_low\_pointwise, red\_low\_integral, red\_high\_integral, normalized\_cdf\_le\_uniform in Upper0423/Uniform/Reduction.lean.

In Lean: norm\_charFun\_sub\_cos\_le\_gamma, norm\_charFun\_le\_cubicExp, low\_geomSum\_le, norm\_charFun\_pow\_sub\_gauss\_le in Upper0423/LargeN/LowPointwise.lean.

## 12.2 The budget

Let  $z = \eta/\beta \in [0, 1]$ , so  $a/\beta = 1 + z$ . Suppose  $F, G, K$  are rational upper bounds for the three bracketed quantities in (22) on a cell  $L \in [L_1, L_2]$ . Two constraints convert (22) into a rational inequality.

**Lemma 12.3** ( $\Gamma$  constraint). *If  $\beta \geq 1$  and  $0 \leq \eta \leq 1$  then  $(\Gamma/\beta)^2 \leq 2(1 - \eta/\beta)$ .*

*Proof.* Equivalently  $(\beta - 1)^2 + \beta^2 - \eta^2 \leq 2\beta^2 - 2\beta\eta$ , i.e.  $(1 - \eta)(2\beta - 1 - \eta) \geq 0$ , which holds since  $\eta \leq 1$  and  $2\beta - 1 - \eta \geq 2 - 1 - 1 = 0$ .  $\square$

**Lemma 12.4** ( $z$  and  $h$  caps). *Let  $r \geq 2$  with  $r^2 \leq n$ , so  $hr \leq 1$ . On a cell with  $L_1 \leq (\beta + \eta)h \leq L_2$ :*

$$z \leq z_{\text{hi}} := \begin{cases} \min(1, \frac{1}{L_1 r - 1}), & L_1 r > 1, \\ 1, & \text{otherwise,} \end{cases} \quad \frac{h}{\beta} \leq \min\left(\frac{1}{r}, \frac{L_2}{2}, \frac{1 + z_{\text{hi}}}{r^2 L_1}\right).$$

*Proof.* From  $L_1 \leq (\beta + \eta)h$  and  $hr \leq 1$  we get  $L_1 r \leq (\beta + \eta)hr \leq \beta + \eta$ , hence  $\beta \geq L_1 r - 1$  (as  $\eta \leq 1$ ) and  $z = \eta/\beta \leq 1/(L_1 r - 1)$ . For the  $h$  caps:  $h \leq 1/r$ ; and  $2h \leq (\beta + \eta)h \leq L_2$  by Proposition 4.3; and  $hr^2 L_1 \leq (hr)^2(\beta + \eta) \leq \beta + \eta \leq (1 + z_{\text{hi}})\beta$ .  $\square$

**Lemma 12.5** (Budget algebra). *Let  $c > 0$ ,  $0 \leq z \leq z_{\text{hi}}$ ,  $0 \leq v$  with  $v^2 \leq 2(1 - z)$ ,  $0 \leq k \leq k_c$ ,  $K \geq 0$ , and  $F' \leq F$ ,  $G' \leq G$ ,  $K' \leq K$ . Then*

$$(1 + z)F' + vG' + kK' \leq F + c + \frac{G^2}{2c} + \max(0, z_{\text{hi}}(F - c)) + k_c K.$$

*Proof.*  $vG \leq \frac{cv^2}{2} + \frac{G^2}{2c}$  (AM-GM, i.e.  $(cv - G)^2 \geq 0$ ) and  $\frac{cv^2}{2} \leq c(1 - z)$ , so

$$(1 + z)F + vG \leq F + zF + c - cz + \frac{G^2}{2c} = F + c + z(F - c) + \frac{G^2}{2c},$$

and  $z(F - c) \leq \max(0, z_{\text{hi}}(F - c))$  in both signs of  $F - c$ .  $\square$

*In Lean:* `bud_gamma_sq`, `bud_z_le`, `bud_hcap`, `bud_kc`, `bud_k_le`, `bud_core`, `bud_val`, `bud_accept`, `bud_accept_sound` in `Upper0423/Uniform/Budget.lean`.

A cell  $[L_1, L_2]$  is therefore *accepted* when  $2 \leq r$ ,  $0 < L_1 \leq L_2 \leq 3$ , the three certified bounds  $F = U_1 + U_2$ ,  $G$ ,  $K$  exist, and

$$F + c + \frac{G^2}{2c} + \max(0, z_{\text{hi}}(F - c)) + k_c K < \frac{423}{1000}$$

for at least one of the three candidate parameters  $c \in \{F, G/\sqrt{2}, G/\sqrt{2(1 - z_{\text{hi}})}\}$  (the square roots replaced by rational under-approximations; the choice of candidate is immaterial to soundness, since *any*  $c > 0$  works in Lemma 12.5).

### 12.3 Certified bounds for the five integrals

It remains to produce  $U_1 \geq (\text{high} + \text{tail})/L$ ,  $U_2 \geq \text{norm}/L$ ,  $G \geq \text{lowG}$  and  $K \geq \text{lowK}$ , uniformly for  $L \in [L_1, L_2]$  and  $h \leq h_{\text{cap}}$ . All four are monotone in the right places, which is what makes a cell-uniform bound possible.

**High and tail.** Both integrands increase with  $L$ , so they are evaluated at  $L_2$ . For the high part,  $\psi$  is bounded below by one of two explicit branches: the supporting line  $\psi(t) \geq (2\pi t)^2(\frac{1}{2} - \kappa 2\pi t)$  (valid for  $2\pi t \leq 2\pi$ ), and the exact curved branch  $\psi(t) = A(1 - \cos 2\pi t)$  once  $2\pi t \geq 4$ . The kernel is bounded by (9) or (8) according to the panel. The resulting smooth majorant is integrated by the panel rule of Lemma 10.6 over  $\lfloor m/3 \rfloor + 1$  panels of  $[\frac{1}{4}, 1]$  and the tail over the remaining panels of  $[\frac{1}{4}, S]$ , plus a closed Gaussian tail beyond  $S$ .

*In Lean:* `ht_psiJet`, `ht_highPanel`, `ht_tailPanel`, `ht_tailRest`, `highTailUpper`, `highTailUpper_sound`, `highTail_small` in `Upper0423/Uniform/HighTail.lean`.

**Normal correction.** Two bounds are computed and the smaller kept. The first is closed-form, from the quadratic kernel majorant  $|K(t) - i/(\pi t)| \leq p(t) = 1 - t + \nu t^2$  on  $(0, \frac{1}{4}]$  (with  $\nu$  as in Table 2) and the exact Gaussian moments:

$$\frac{\text{norm}(L)}{L} \leq \frac{\sqrt{\pi/2}}{2\pi} - \frac{L_1}{4\pi^2} + \frac{\nu\sqrt{\pi/2}L_2^2}{8\pi^3}.$$

The second is a left-rectangle sum in the variable  $v = Tt$ : on the  $t$ -panel corresponding to  $v \in [ih, (i+1)h]$  one has  $p(t) \leq p(L_1 ih/(2\pi))$  and  $e^{-T^2 t^2/2} \leq e^{-(ih)^2/2}$ , and the panel width is at most  $Lh/(2\pi)$ , giving

$$\frac{\text{norm}(L)}{L} \leq \frac{h}{2\pi} \sum_{i < m} p\left(\frac{L_1 ih}{2\pi}\right) e^{-(ih)^2/2}.$$

The sum is evaluated in interval arithmetic by a recursion that carries  $e^{-(ih)^2/2} = x^{i^2}$  with  $x = e^{-h^2/2}$ , so that only one exponential is ever computed.

*In Lean:* `nb_step, nb_loop, nb_rect, nb_closed, normalUpper, normalUpper_sound, normalIntegral_le, gaussMoment_*` in `Upper0423/Uniform/NormalBound.lean`, `Upper0423/LargeN/Normal.lean`.

**The low coefficients.** Substituting  $v = 2\pi t/L$  turns `lowG` into  $\int_0^{\pi/(2L)} s|K(s)| \frac{v^2}{12} (E_1 + E_2) dv$  with  $s = Lv/(2\pi)$ , and similarly for `lowK`. On each  $v$ -panel  $[\alpha, \beta'] \subseteq [0, V]$ :

- the kernel factor  $s|K(s)|$  is bounded by a prefix maximum over a precomputed table on the grid  $s = j/(4N)$ ,  $N = 1024$ ; each table entry uses (9) in the form  $s^2|K(s)|^2 = \sigma^2 + (\frac{1}{\pi} - \sigma q(\pi s))^2$  with  $\sigma = s(1 - s)$ , both factors being monotone on the relevant range;
- $E_1 \leq e^{-\varrho v^2/2}$  and  $E_2 \leq e^{-\varrho v^2/2 + \varrho \kappa L_e v^3}$  with  $\varrho = 1 - h_{\text{cap}}^2$  and  $L_e = \min(L_2, \pi/(2\alpha))$ , the second using  $v \leq \pi/(2L)$ ;
- the resulting smooth majorant is integrated by the fourth-order panel rule of Lemma 10.6.

When  $V = 8 < \pi/(2L)$  a closed-form Gaussian tail is added beyond 8.

*In Lean:* `lb_table, lb_panel, lb_tail, lowUpper, lb_panel_sound, lb_tail_sound, lowUpper_sound` in `Upper0423/Uniform/LowBound.lean`.

## 12.4 The cells

**Theorem 12.6.** *Let  $r = 16$ ,  $m = 256$ . There is a list of 282 rational edges*

$$\frac{1}{100} = e_0 < e_1 < \dots < e_{281} = \frac{260889}{100000}$$

*such that every consecutive cell  $[e_i, e_{i+1}]$  is accepted, and such that the initial cell  $(0, \frac{1}{100}]$  is accepted in the variant form in which  $(\text{high} + \text{tail})/L \leq 1/1000$  is proved analytically. Consequently, for every admissible  $\mu$ , every  $n \geq 256$  with  $\beta/\sqrt{n} \leq \frac{13003}{10000}$ , and every  $x$ ,*

$$\mathcal{E}(\mu, n, x) \leq \frac{423}{1000}.$$

*In Lean:* `cellAccepted, cellsAccepted, smallCellAccepted, bud_cover, normalizedError_le_of_cellsAccepted` in `Upper0423/Uniform/Budget.lean`.

*In Lean:* `chunk0 ... chunk8, acc0 ... acc8, normalizedError_uniform` in `Upper0423/Uniform/Data/Chunk0.lean ... Chunk8.lean, Uniform/Data/All.lean`.

*Proof of the deduction.* Set  $L = (\beta + \eta)h$  with  $h = 1/\sqrt{n}$ . Since  $n \geq 256 = r^2$  we have  $r \leq \sqrt{n}$ , so  $hr \leq 1$ . The upper end:  $\eta \leq \beta$  (from  $\eta \leq 1 \leq \beta$ ) gives  $L \leq 2\beta/\sqrt{n} \leq 2 \cdot \frac{13003}{10000} = \frac{26006}{10000}$ , which is at most the last edge  $e_{281} = 2.60889$ . If  $L \leq \frac{1}{100}$  the initial cell applies; otherwise  $L$  lies in one of the 281 cells, found by a linear scan (`bud_cover`). Proposition 12.2, the certified bounds  $F, G, K$  for that cell, and Lemmas 12.3–12.5 give  $\mathcal{E}(\mu, n, x) < \frac{423}{1000}$ .  $\square$

The 281 cells are declared in nine files `Uniform/Data/Chunk $k$`  (`Chunk0–Chunk7` with 35 cells each, `Chunk8` with one), each discharged by a single `native_decide`; consecutive chunks share an edge and are glued by `cellsAccepted_append`. A tenth `native_decide`, in `Uniform/Data/All.lean`, discharges `smallCellAccepted` 16 256. The edges are geometric with ratio  $\approx 1.02$  (the generator was invoked as `gen_uniform.py 256 102/100 35`).

**Remark 12.7.** The cell widths matter: the certified  $F$ ,  $G$ ,  $K$  are constant on a cell but the true integrals vary, so a cell that is too wide is rejected, while too many cells make the `native_decide` evaluation slow. The interesting statistic is the largest certified cell value, i.e. the left-hand side of the budget inequality maximized over the cells. Evaluating the same Lean functions `highTailUpper`, `normalUpper`, `lowUpper` and `bud_val` on all 281 cells gives

$$\max_{1 \leq i \leq 281} (\text{budget value of the cell } [e_{i-1}, e_i]) = 0.414962\dots,$$

and the initial cell  $(0, \frac{1}{100}]$  gives  $0.412306\dots$ , so the certificate clears  $423/1000$  with a margin of about  $0.008$ . (This is an auxiliary evaluation of the same definitions, reported for orientation; it is not part of the formal proof, which needs only the strict cellwise inequality that `native_decide` establishes. The figure agrees with the approximate value  $0.4150$  recorded in `docs/UPPER_0423.md`.)

## 13 Trust boundary

### 13.1 Axioms

Running

```
#print axioms BerryEsseen.Upper0423.berryEsseenConstant_le_0423
```

reports

- the three standard axioms of Lean’s type theory: `propext`, `Classical.choice`, `Quot.sound`; and
- 579 auxiliary axioms generated by `native_decide`, one per finite certificate.

We verified the count 579 independently by enumerating the `native_decide` invocations in the module tree: 507 in `Covers/Data/` (one per cover file: 253 BE files and 254 DW files), 62 in `Small/Data/` (one per cover file), 9 in `Uniform/Data/Chunkk` and 1 in `Uniform/Data/All.lean`;  $507 + 62 + 9 + 1 = 579$ .

There is no `sorry` anywhere in the development and no user-declared axiom. The consequence of the 579 auxiliary axioms is precise and limited: for each of the 579 closed Boolean expressions, the proof asserts that the expression evaluates to `true`, on the evidence of Lean’s compiler and runtime rather than of Lean’s kernel. The mathematical content — the smoothing inequality, all the characteristic-function estimates, the semantics of the interval arithmetic and the jets, the panel rule, and every soundness theorem relating a Boolean acceptance predicate to a real-analytic statement — is kernel-checked.

### 13.2 What is and is not claimed

- We claim that the Lean development in `BerryEsseen/Upper0423/` builds and that its final theorem is the statement displayed in Section 3, with the axiom profile just described.
- We claim no independent verification of the development beyond the Lean build and the axiom query described above.
- We claim no peer review of this document or of the development.
- We make no claim about the correctness of [16] or of [2].
- The definition of the constant being bounded is the one displayed in Section 3. Readers who wish to audit only the statement need read only `Challenge.lean`, which imports `Mathlib` alone.

### 13.3 Residual risks

The honest list of things that could still be wrong is:

- (1) a bug in Lean’s compiler or runtime affecting the evaluation of one of the 579 Boolean expressions (mitigated by the fact that the expressions use only arbitrary-precision integer and rational arithmetic, lists, arrays and structural recursion);
- (2) a bug in Lean’s kernel or in Mathlib (the standard risk of any formalization);
- (3) a mismatch between the definition of `berryEsseenConstant` and the informal statement of Theorem 1.1 (mitigated by the verbatim reproduction in Section 3);
- (4) an error in this document’s *description* of the Lean proof. The Lean sources, not this document, are authoritative; every lemma above carries the name and file of its formal counterpart so that the two can be compared.

## 14 Reproduction

The development is built with

```
lake build BerryEsseen.Upper0423.Main
```

The default library target `BerryEsseen` imports it, so `lake build BerryEsseen` builds it as well. A full build is dominated by the evaluation of the 579 certificates.

The Lean data files are *generated*, by the scripts in `scripts/upper_0423/`:

- `run_covers.sh` (with `make_jobs.py`, `cover_gen.py`) searches the covers using the research producer `research/native_cover.py` and its C++ engine;
- `gen_covers.py`, `gen_all.py` write `Covers/Data/*.lean` and `Covers/All.lean`;
- `gen_small_cover.py`, `gen_small_data.py` write `Small/Data/*.lean` from the research covers;
- `gen_uniform.py` 256 102/100 35 writes `Uniform/Data/*.lean`.

The search is entirely untrusted: it chooses the tree shapes, the smoothing parameters  $c$  and  $\tau$ , the panel subdivisions and the branch choices, but every resulting numerical bound is recomputed from scratch by the Lean checkers, whose soundness is proved for all accepted inputs. In particular a faulty search can only produce a rejected certificate, never a false theorem.

## 15 Related and concurrent work

**Xiao–Li.** The repository [16] states the bound  $C_{\text{BE}} \leq 879/2000 = 0.4395$  with a Lean 4 formalization, also using `native_decide` for its certificates. Its analytic framework is a direct ancestor of the present one; see Section 2 for the itemized attribution. As stated there, neither its numerical result nor its certificates nor its Lean code are used in Theorem 1.1, and we make no assessment of its correctness. It is an unpublished repository, with an accompanying note, not a peer-reviewed publication.

**The independent-summand constant.** The repository [2] treats the analogous constant  $C_{\text{ind}}$  for independent, not necessarily identically distributed, summands, and states  $C_{\text{E}} \leq C_{\text{ind}} < 0.44995$  along with several related results. This is a different constant (necessarily  $C_{\text{ind}} \geq C_{\text{BE}}$ ), and nothing from that collection is used here. It is likewise an unpublished collection of preprints, and we make no assessment of its correctness.

**Relation to the published record.** Against Table 1, Theorem 1.1 would improve the best published bound 0.4690 of [9]. We stress again that we neither use nor reprove any of the entries of that table, and that the present document is not peer reviewed.

**Relation to an earlier formalization.** The repository previously contained a formalization of the weaker bound  $C_{\text{BE}} \leq 0.4688$  that avoided `native_decide`. That upper-bound proof has been removed; it remains available in the repository’s Git history. The 0.423 development reuses general-purpose modules originally written for it (the definitions, the Prawitz smoothing inequality and its integrability, kernel estimates, characteristic-function anchors and the Cantelli bound), which remain in the repository together with the formalized lower bound  $0.40 \leq C_{\text{BE}}$ .

## Acknowledgements

The analytic framework builds on the work of Haonan Xiao and Chunlin Li [16] as detailed in Section 2. The formalization rests entirely on Mathlib [13]. The historical data of Table 1 follow the survey entry [12].

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